# Numerical analysis of superconducting phases in the extended Hubbard model with non-local pairing

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**Abstract**

[To be continued...]

- Appendix on $s$-wave SC ghost simulations on HM
- Appendix on triplet pairing in MFT self-consistent scheme
- Set ComputerModern font in all CairoMakie figures

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**List of symbols and abbreviations**

| | |
|---|---|
| AF | Antiferromagnetic |
| BCS | Bardeen-Cooper-Schrieffer (theory) |
| BZ | Brillouin zone |
| $\beta$ | Inverse temperature |
| $\Delta^{(\gamma)}$ | $\gamma$-wave component of the gap function |
| DoF | Degree of freedom |
| EHM | Extended Hubbard model |
| EV(s) | Expectation value(s) |
| FS | Fermi surface |
| HF | Hartree-Fock |
| HFP(s) | Hartree-Fock parameter(s) |
| HM | Hubbard model |
| $L_\ell$ | Number of lattice sites in direction $\ell$ |
| iHF | Iterative Hartree-Fock |
| LRT | Linear Response Theory |
| MBZ | Magnetic Brillouin zone |
| MFT | Mean-Field Theory |
| NBZ | Nematic Brillouin zone |
| NN(s) | Nearest neighbor(s) |
| PH | Particle-hole |
| PHS | Particle-hole symmetry |
| RWC(s) | Relevant Wick's contraction(s) |
| RMP(s) | Renormalized model parameter(s) |
| SC | Superconductor |
| SDW | Spin-density wave |
| SSB | Spontaneous symmetry breaking |
| $T_\mathrm{c}$ | Critical temperature |

# Introduction

This thesis project is about my favorite ice cream flavor. [To be continued. . . ]

# Chapter 1

# Introduction to the Extended Hubbard Model

A major breakthrough in Condensed Matter Physics was the discovery of superconductivity in two-dimensional copper-oxide based compounds (often referred to as "cuprates"). Until 1986, when Bednorz and Muller found evidence of a superconducting transition in Ba-La-Cu-O systems [3], superconductivity had been observed up to critical temperatures ($T_\mathrm{c}$) around 30 K. The new class of copper oxides rapidly took over as highest-$T_\mathrm{c}$ class of superconductors, dramatically increasing critical temperatures to around 130 K at ambient pressure. Today, the highest observed value is $T_\mathrm{c} \approx 151$ K at ambient pressure on the cuprate $\mathrm{HgBa_2Ca_2Cu_3O_{8+\delta}}$ [4]. It is worth mentioning that this result was obtained in recent months after almost 30 years from the previous record.

Cuprates Physics is particularly interesting and surprising also considering their very poor conducting properties at normal phase. Copper oxides are basically ceramics, highly prone to establishing insulating antiferromagnetic phases. Antiferromagnetism and superconductivity are apparently antithetical orderings: the first arises from electronic repulsion, while the second from attractive interactions [5]. It is widely observed, however, how in cuprates the two phases can compete **and even coexist** [6].

Moreover, cuprates superconduct through a new, not-yet understood mechanism. Classical Bardeen-Cooper-Shrieffer (BCS) theory of superconductivity describes the arising of a phonon-induced effective electron-electron attractive interaction as a source of so-called "conventional superconductivity". It is the interplay of electrons with lattice vibrations that couples them in Cooper pairs. The same theory fails completely in explaining this new "unconventional superconductivity". All of these observation have gathered, in years, a lot of interest in these materials.

Such complex materials are surprisingly well described in many features by the simple (but rich) single-band Hubbard model (HM) [6],

$$\hat{H}_{\mathrm{HM}} = -t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + U \sum_{i} \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} \qquad t, U > 0 \tag{1.1}$$

being $t$ the hopping energy and $U$ the on-site repulsion. Sum is intended on the spin index $\sigma \in \{\uparrow, \downarrow\}$ and the sites $i$ of a two-dimensional square lattice. The notation $\langle ij \rangle$ indicates nearest neighbours (NNs). Recent developments [7] have shown, however, that a more accurate description of these materials could be obtained by adding a non-local electron-electron attraction. This leads us to a widely studied modified version of Eq. (1.1),

$$\hat{H}_{\mathrm{EHM}} = -t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{c}^{\dagger}_{i\sigma} \hat{c}_{j\sigma} + U \sum_{i} \hat{n}_{i\uparrow} \hat{n}_{i\downarrow} - V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{n}_{i\sigma} \hat{n}_{j\sigma'} \qquad t, U, V > 0 \tag{1.2}$$

commonly referred to as "Extended Hubbard model" (EHM). The difference with the conventional HM is the presence of a NNs attractive interaction, independent of site and spin indices. In this thesis, we study the insurgence of antiferromagnetism and superconductivity in the EHM due to the presence of this new interaction. To do so, we use numerical Hartree-Fock (HF) methods.

This introductory chapter is organised as follows: [To be continued...]

# Chapter 2

# Introduction to Hartree-Fock methods

The Hartree-Fock (HF) method is a variational approach to determine the ground-state wavefunction of a many-body hamiltonian. The wavefunction is determined the one minimising energy, limiting search to a specific subspace of states. The general idea is to approximate the real hamiltonian with the best non-interacting fermionic hamiltonian.

To start, we need to define what Hilbert subspace we work with. Then we need to identify a way of parametrising the possible non-interacting hamiltonians and their respective ground-states, and finally a method to choose the best approximation to our original model. This is done in Sec. 2.2, largely based on [26, 28].

Variational algorithms generally work by numeric minimisation of some given functional. This is not the case. We implement HF theory starting from a theoretical minimisation procedure, with some derivative being set to zero, and then prove that this procedure leads to solving a self-consistency problem. We end up with a set of equations where the right-hand side depends non-trivially on the left-hand side, and through an iterative approach we determine the solution. This procedure is sketched in detail in Sec. 2.3.

## 2.1 Prerequisites and generalities

Let us start by discussing the properties of the many-body states we use to approximate the real ground-state of the system. We introduce Slater determinants to grasp the general idea behind HF variational method, and then extend to a larger set of states.

### 2.1.1 Slater determinants

A many-body wavefunction describing a collective electronic state needs to be antisymmetric under interchange of coordinates and spins $(\mathbf{r}_i, \sigma_i)$, $(\mathbf{r}_j, \sigma_j)$ of any couple of electrons. The simplest way to build a many-body wavefunction satisfying this rule is "to antisymmetrize" a set of orthonormal $N$ one-body spin-orbitals, being $N$ the total number of electrons and $D$ the total number of spin-orbitals. Let $\{\psi_i\}$ be said set,

$$\frac{1}{2\mathcal{V}} \sum_{\sigma \in \{\uparrow, \downarrow\}} \int \mathrm{d}\mathbf{r}\ \psi_i(\mathbf{r}, \sigma) \psi_j(\mathbf{r}, \sigma) = \delta_{ij}$$

being $\mathcal{V}$ the total volume.

Let $\mathrm{S}_N$ be the $N!$ elements symmetric group of permutations of $N$ objects and $\mathrm{P} \in \mathrm{S}_N$ a specific permutation of a given $N$-uple,

$$\mathrm{P} \quad : \quad (x_1, x_2, \cdots, x_N) \to \left(x_{\mathrm{P}(1)}, x_{\mathrm{P}(2)}, \cdots, x_{\mathrm{P}(N)}\right)$$

Let $F_{\mathrm{P}}$ be the operator that implements the permutation P on the $N$ variables of a function $f$,

$$F_{\mathrm{P}}\left[f(x_1, x_2, \cdots, x_N)\right] = f\left(x_{\mathrm{P}(1)}, x_{\mathrm{P}(2)}, \cdots, x_{\mathrm{P}(N)}\right)$$

We define the antisymmetrization operator:

$$\mathcal{A}[f] \equiv \frac{1}{\sqrt{N!}} \sum_{\mathrm{P} \in \mathrm{S}_N} (-1)^{\sigma(\mathrm{P})} F_{\mathrm{P}}[f]$$

being $\sigma(\mathrm{P})$ the "parity factor" of the permutation $\mathrm{P} \in \mathcal{S}_N$. The factor $\sigma(\mathrm{P})$ counts the number of exchanges needed to obtain P: this way, an even number of exchanges gives a total sign $+1$, an odd number gives $-1$. By construction, the function $\mathcal{A}[f]$ is antisymmetric under exchange of a pair of coordinates.

Consider now the many-body wavefunction

$$\Psi(\mathbf{r}_1, \sigma_1; \mathbf{r}_2, \sigma_2; \cdots; \mathbf{r}_N, \sigma_N) = \mathcal{A}\left[\prod_{i=1}^{N} \psi_i(\mathbf{r}_i, \sigma_i)\right]$$

This class of wavefunctions, obtained as antisymmetric combinations of product states, are representable as so-called Slater determinants,

$$\Psi(\mathbf{r}_1, \sigma_1; \mathbf{r}_2, \sigma_2; \cdots; \mathbf{r}_N, \sigma_N) = \frac{1}{\sqrt{N!}} \det \begin{bmatrix} \psi_1(\mathbf{r}_1, \sigma_1) & \psi_1(\mathbf{r}_2, \sigma_2) & \cdots & \psi_1(\mathbf{r}_N, \sigma_N) \\ \psi_2(\mathbf{r}_1, \sigma_1) & \psi_2(\mathbf{r}_2, \sigma_2) & \cdots & \psi_2(\mathbf{r}_N, \sigma_N) \\ \vdots & \vdots & & \vdots \\ \psi_N(\mathbf{r}_1, \sigma_1) & \psi_N(\mathbf{r}_2, \sigma_2) & \cdots & \psi_N(\mathbf{r}_N, \sigma_N) \end{bmatrix}$$

This representation includes automatically Pauli exclusion principle: determinants of matrices with two equal rows or two equal columns are null. Thus, all spin-orbitals must differ, as well as no single spin-orbital $\psi_i$ can be doubly occupied (which is, all spin-coordinate couples must be different).

Any normalised combinations of Slater determinants gives an antisymmetric wavefunction, although not necessarily representable as a single Slater determinant. In HF theory, we search for the best possible choice of orthonormal one-body spin orbitals $\{\psi_1, \psi_2, \cdots, \psi_N\}$ to approximate the true ground-state of the model through a single Slater determinant. The space of possible "Slater-like" many-body wavefunctions is the parametric space where we perform the variational constrained search.

The EHM is formulated in second-quantisation. It is then useful to represent Slater-like states in this language. Let us map all the spin-orbitals from the discussion above to pairs of annihilation-creation fermionic operators,

$$\psi_i \quad \rightarrow \quad \hat{c}_i, \hat{c}_i^\dagger$$

Any Slater state can be expressed in second-quantisation as:

$$|\Psi\rangle = \prod_{i=1}^{N} \hat{c}_i^\dagger |\Omega\rangle \tag{2.1}$$

being $|\Omega\rangle$ the vacuum. Antisymmetry is guaranteed by Fermi anticommutation rules.

### 2.1.2 Wick's theorem applied to Slater determinants

The concept of vacuum state needs to be interpreted in relation to an appropriate choice of second-quantisation annihilation and creation operators. For a given set of fermion operators, the vacuum is that state mapped to zero by any annihilation operator. As we will see, to correctly define the vacuum state is crucial in order to apply Wick's theorem and develop an HF approximation of EHM. Before moving to generalisation, let us discuss a playground setup.

Consider some unspecified model whose ground state at null temperature is given in Eq. (2.1) for a given set of electronic operators $\hat{c}_i$, $\hat{c}_i^\dagger$. Let $\hat{c}_i$ with $N < i < D$ identify all spin-orbitals not already included in the definition of $|\Psi\rangle$ of Eq. (2.1) (thus, non populated). Then it holds:

$$\begin{aligned} 0 \leq i \leq N \qquad & \hat{c}_i^\dagger |\Psi\rangle = 0 \\ i > N \qquad & \hat{c}_i |\Psi\rangle = 0 \end{aligned}$$

Let:

$$\hat{a}_i \equiv \begin{cases} \hat{c}_i^\dagger & \text{for } 0 \leq i \leq N \\ \hat{c}_i & \text{for } i > N \end{cases}$$

These new operators satisfy a Fermi algebra $\{\hat{a}_i, \hat{a}_j^\dagger\} = \delta_{ij}$ and annihilate the "vacuum" state $|\Psi\rangle$. In other words, any Slater state like Eq. (2.1) can be seen as a vacuum state for a suitably chosen set of fermionic operators.

This property is handy for evaluation of expectation values (EV). Let $\hat{O}^{(\ell)}$ be a linear combination of $\hat{c}_i$ or $\hat{c}_i^\dagger$,

$$\hat{O}^{(\ell)} \equiv \sum_{i=1}^{D} \left( u_i^{(\ell)} \hat{c}_i + v_i^{(\ell)} \hat{c}_i^\dagger \right) \qquad u_i^{(\ell)}, v_i^{(\ell)} \in \mathbb{C} \tag{2.2}$$

Consider the product operator:

$$\hat{\mathcal{O}} \equiv \hat{O}^{(1)} \hat{O}^{(2)} \cdots \hat{O}^{(2k)} \qquad k \in \mathbb{N}$$

Inserting the above decomposition, we get a large sum of many terms, each containing some sequence of annihilations and creations. Using linearity we can concentrate on one specific term of these, derive results and then recompose them for the original operator. Consider one of these terms,

$$\hat{\mathcal{O}} = \cdots + (\text{coefficient}) \times \hat{C}_1 \hat{C}_2 \cdots \hat{C}_{2k} + \cdots$$

with $\hat{C}_i$ some fermionic operator, either $\hat{c}_j$ or $\hat{c}_j^\dagger$ for some $j$. Now, Wick's theorem states that

$$\begin{aligned} \hat{C}_1 \hat{C}_2 \cdots \hat{C}_{2k} = {} & \mathrm{N}\left\{ \hat{C}_1 \hat{C}_2 \cdots \hat{C}_{2k} \right\} \\ & + \sum_{(\alpha\beta)} (-1)^{\zeta_{\alpha\beta}} D_{\alpha\beta} \mathrm{N} \left\{ \prod_{i \neq \alpha,\beta} \hat{C}_i \right\} \\ & + \sum_{(\alpha\beta)(\gamma\delta)} (-1)^{\zeta_{\alpha\beta}} D_{\alpha\beta} (-1)^{\zeta_{\gamma\delta}} D_{\gamma\delta} \mathrm{N} \left\{ \prod_{i \neq \alpha,\beta,\gamma,\delta} \hat{C}_i \right\} \\ & + \cdots \end{aligned}$$

where $\mathrm{N}\{\cdots\}$ indicates normal ordering, $D_{\alpha\beta}$ is the contraction of the pair $\hat{C}_\alpha \hat{C}_\beta$ and $\zeta_{\alpha\beta}$ is a symmetry factor included to take care of the number of fermionic algebra. In this context contractions are just expectation values over $|\Psi\rangle$. Sums are performed over pairs of indices, implicitly taken to be different from each other. To take averages of this last operator over $|\Psi\rangle$ is a not-so-hard task, given the simple expression of the state in terms of electronic annihilations and creations. However it can be made simpler: if we change basis and express all operators in terms of $\hat{a}$, $\hat{a}^\dagger$,

$$\hat{C}_i \quad \rightarrow \quad \hat{A}_i$$

the above decomposition holds identically, with slight change of notation. Now, taking a zero temperature average on $|\Psi\rangle$ (the vacuum with respect to the new operators) all normal-ordered terms disappear,

$$\langle \Psi | \mathrm{N}\{\hat{A}_i \hat{A}_j \cdots\} | \Psi \rangle = 0$$

thus in the decomposition above only fully contracted terms survive. Going back to $\hat{c}$ operators, we can re-express fully contracted terms in the original basis. Thus:

$$\left\langle \hat{C}_1 \hat{C}_2 \cdots \hat{C}_{2k} \right\rangle = \sum (\text{total sign factor}) \times (\text{product of complete contractions})$$

where sum is intended, implicitly, over all possible ways of choosing $k$ couples of indices from a starting set of $2k$ elements.

Taking another step back, due to linearity we can apply this result to all terms of $\mathcal{O}$ and conclude that the EV of $\mathcal{O}$ decomposes as a sum of 2-points EVs. **This is the defining property**

**of gaussian states**. In this procedure, the only key property of Slater states we used was the fact that they are vacuum to some fermionic basis. Finite-temperature extension of Wick's theorem as well as more complicated change of basis are possible, but the concept remains the same: on a Slater state, which is essentially a non-interacting state for some set of fermions, Wick's theorem is particularly well-behaved.

From the point above we can conclude that:

$$\langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta \rangle = \langle \hat{c}^\dagger_\alpha \hat{c}_\delta \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\gamma \rangle - \langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle \langle \hat{c}^\dagger_\gamma \hat{c}_\delta \rangle$$

being the average intended on a Slater state. In this context, only contractions with an equal number of creation and annihilation operators contribute.

### 2.1.3 Generalisation to Bogoliubov-Valatin states

In HF theory we look for a good approximation of the true many-body wavefunction constraining search to a class of states. Introducing Slater determinants was useful to grasp the concrete idea behind the method. **However, in order to treat some physical phenomena we need to extend the variational procedure to a other sets of states. For example, Bardeen-Cooper-Shrieffer (BCS) theory of superconductivity gives a many-body ground-state which is not representable as a Slater determinant of "standard" electronic operators.** In this section we extend the notions introduced before and set grounds for the generalised Hartree-Fock-Bogoliubov (HFB) method we use in this thesis.

Consider the Nambu spinor

$$\hat{\Phi} \equiv \begin{bmatrix} \hat{\mathbf{c}} \\ \hat{\mathbf{c}}^\dagger \end{bmatrix}$$

being $\hat{\mathbf{c}}$ the vector of all annihilation operators, ordered, and $\hat{\mathbf{c}}^\dagger$ of all creation operators. Let us assume the vectors have $D$ entries. Consider an unitary map $M$, such that

$$\hat{\Gamma} \equiv M\hat{\Phi} \quad \text{being} \quad \hat{\Gamma} \equiv \begin{bmatrix} \hat{\boldsymbol{\gamma}} \\ \hat{\boldsymbol{\gamma}}^\dagger \end{bmatrix}$$

The new Nambu vectors $\hat{\boldsymbol{\gamma}}$ and $\hat{\boldsymbol{\gamma}}^\dagger$ contain our new, generalised fermionic operators in second quantisation. Let:

$$M \equiv \begin{bmatrix} U & V \\ U^\dagger & V^\dagger \end{bmatrix}$$

If $M$ is unitary, it follows:

$$\begin{aligned}
\mathbb{I}_{2D\times 2D} &= MM^\dagger \\
&= \begin{bmatrix} U & V \\ U^\dagger & V^\dagger \end{bmatrix} \begin{bmatrix} U & U^\dagger \\ V^\dagger & V \end{bmatrix} \\
&= \begin{bmatrix} UU^\dagger + VV^\dagger & U^2 + V^2 \\ \left(U^\dagger\right)^2 + \left(V^\dagger\right)^2 & U^\dagger U + V^\dagger V \end{bmatrix}
\end{aligned}$$

thus, among other conditions, $UU^\dagger + VV^\dagger = \mathbb{I}_{D\times D}$.

The new operators in $\hat{\Gamma}$ satisfy Fermi anticommutation rules. In fact, writing the mapping explicitly for any of the new operators and some index $i \in [1, D]$,

$$\hat{\gamma}_i \equiv \sum_{j=1}^{D} U_{ij}\hat{c}_j + \sum_{j=1}^{D} V_{ij}\hat{c}^\dagger_j \qquad \hat{\gamma}^\dagger_i \equiv \sum_{j=1}^{D} U^*_{ji}\hat{c}^\dagger_j + \sum_{j=1}^{D} V^*_{ji}\hat{c}_j$$

and substituting in the anti-commutator

$$\begin{aligned}
\{\hat{\gamma}_i, \hat{\gamma}_j^\dagger\} &= \sum_{i'=1}^{D} U_{ii'} \sum_{j'=1}^{D} U^*_{j'j} \{\hat{c}_{i'}, \hat{c}^\dagger_{j'}\} + \sum_{i'=1}^{D} V_{ii'} \sum_{j'=1}^{D} V^*_{j'j} \{\hat{c}^\dagger_{i'}, \hat{c}_{j'}\} \\
&= \sum_{i'=1}^{D} U_{ii'} \sum_{j'=1}^{D} U^*_{j'j} \delta_{i'j'} + \sum_{i'=1}^{D} V_{ii'} \sum_{j'=1}^{D} V^*_{j'j} \delta_{i'j'} \\
&= \left[UU^\dagger + VV^\dagger\right]_{ij} \\
&= \delta_{ij}
\end{aligned}$$

being $M$ unitary.

Assume now $|\Psi\rangle$ to be the vacuum state for the new operators,

$$\forall i \quad : \quad \hat{\gamma}_i |\Psi\rangle = 0$$

Suppose we want to evaluate, with identical meaning of the notation as in previous section, something like

$$\left\langle \hat{C}_1 \hat{C}_2 \cdots \hat{C}_{2k} \right\rangle$$

We can apply the result obtained above. Indeed, inverse maps linking old and new operators read:

$$\hat{c}_i \equiv \sum_{j=1}^{D} U_{ij} \hat{\gamma}_j + \sum_{j=1}^{D} U^*_{ji} \hat{\gamma}^\dagger_j \qquad \hat{c}^\dagger_i \equiv \sum_{j=1}^{D} V^*_{ji} \hat{\gamma}^\dagger_j + \sum_{j=1}^{D} V_{ij} \hat{\gamma}_j$$

In Sec. 2.1.2 we argued that, when averaging a product of operators like the one of Eq. (2.2), only 2-points EVs contribute. This time the idea is the same, substituting $\mathcal{O}^{(\ell)} \to \hat{C}_\ell$ and $\hat{c}_i, \hat{c}_i^\dagger \to \hat{\gamma}_i, \hat{\gamma}_i^\dagger$. Thus, also here we can decompose the product operator, apply Wick's theorem, discard all non-fully-contracted terms and recompose the results. All we used was just the unitary mapping between $\hat{\Phi}$ and $\hat{\Gamma}$, the fact that $\gamma_i, \hat{\gamma}_i^\dagger$ obey Fermi anticommutation rules and "see" $|\Psi\rangle$ as a vacuum state.

As in Sec. 2.1.2, we conclude that:

$$\langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta \rangle = \underbrace{\langle \hat{c}^\dagger_\alpha \hat{c}_\delta \rangle \langle \hat{c}^\dagger_\beta \hat{c}_\gamma \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle \langle \hat{c}^\dagger_\gamma \hat{c}_\delta \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \rangle \langle \hat{c}_\gamma \hat{c}_\delta \rangle}_{\text{Cooper}} \tag{2.3}$$

The subscripts indicate the name we will give to these terms throughout the text. Note that, differently from "pure Slater states" EVs with two annihilations or two creations do not vanish, in general.

All of this holds when averaging is performed on a state, $|\Psi\rangle$, which is vacuum for a set of operators linked to fermionic Nambu spinor $\hat{\Phi}$ by some unitary operation. This set of states is the domain of our variational constrained search. This kind of unitary mapping is often referred to as Bogoliubov-Valatin transformation, and the method hereby presented as Hartree-Fock-Bogoliubov (HFB). For simplicity we will stick, from now on, to the name "HF method" referring to this discussion. **This argument has been carried out at zero temperature, but can be extended to finite temperature using appropriate path-integral techniques**.

## 2.2 Sketch of HF variational method

We have identified the features of the states we use to approximate the model ground-state. Now, let us get to the core of HF method and give a technical introduction on how it works. Consider the generic quartic hamiltonian:

$$\hat{H} = \sum_{\alpha,\beta} A_{\alpha\beta} \hat{c}^\dagger_\alpha \hat{c}_\beta + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta} \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta$$

where

$$A_{\alpha\beta} = A^*_{\beta\alpha} \qquad B_{\alpha\beta\gamma\delta} = B^*_{\delta\gamma\beta\alpha} \tag{2.4}$$

to ensure hermiticity. Moreover, the model must show the following symmetry:

$$B_{\alpha\beta\gamma\delta} = B_{\beta\alpha\delta\gamma} \tag{2.5}$$

because exchanging the position of the two annihilations and the two creations, the operator must remain the same.

Our aim is to approximate the above hamiltonian with something like:

$$\hat{H} \simeq \hat{H}_{\mathrm{HF}} \equiv \sum_{\alpha,\beta} k_{\alpha\beta}\hat{\gamma}^\dagger_\alpha \hat{\gamma}_\beta \tag{2.6}$$

Old and new operators must be linked by an unitary map: with identical notation as in previous sections, $\hat{\Gamma} = M\hat{\Phi}$. To find the best approximation to the original hamiltonian means to identify both the $\gamma$ operators and the $k$ coefficients. Let us discuss briefly the features of $\hat{H}_{\mathrm{HF}}$ and then propose a solution.

### 2.2.1 Properties of the target quadratic hamiltonian

In this short section we enumerate some useful properties of $\hat{H}_{\mathrm{HF}}$. We will use also the following notation:

$$\hat{a}^-_\alpha \equiv \hat{c}_\alpha \qquad \hat{a}^+_\alpha \equiv \hat{c}^\dagger_\alpha$$

Due to linearity of the unitary mapping, the operator $\hat{H}_{\mathrm{HF}}$ of Eq. (2.6) can also be expressed as the most general quadratic hamiltonians for electronic operators:

$$\hat{H}_{\mathrm{HF}} = \sum_{\alpha,\beta}\sum_{p,q} h^{pq}_{\alpha\beta}\hat{a}^p_\alpha \hat{a}^q_\beta \tag{2.7}$$

By changing basis from $\hat{\gamma}$ to $\hat{c}$ operators, we moved all our ignorance into the new $h^{pq}_{\alpha\beta}$ fields.

We can infer some mathematical properties of the fields $h^{pq}_{\alpha\beta}$. First, note that the hamiltonian can be represented through Nambu spinors as

$$\hat{H}_{\mathrm{HF}} = \begin{bmatrix} \hat{\mathbf{c}} \\ \hat{\mathbf{c}}^\dagger \end{bmatrix}^\dagger \begin{bmatrix} h^{+-} & h^{++} \\ h^{--} & h^{-+} \end{bmatrix} \begin{bmatrix} \hat{\mathbf{c}} \\ \hat{\mathbf{c}}^\dagger \end{bmatrix}$$

Our hamiltonian must be hermitian. We conclude that:

$$\hat{H}^\dagger_{\mathrm{HF}} = \hat{H}_{\mathrm{HF}} \qquad \Longrightarrow \qquad h^{++} = \left[h^{--}\right]^\dagger$$

which reads, element-wise:

$$h^{++}_{\alpha\beta} = \left(h^{--}_{\beta\alpha}\right)^* \tag{2.8}$$

Same-sign off-diagonal fields are linked.

Moreover, any hamiltonian can be arbitrarily chosen to be traceless: Physics is not affected by constant energy shifts. Indeed, if an hamiltonian is not traceless, it suffices to subtract its trace to get a traceless and physically equivalent operator. Thus we can set the target hamiltonian to be traceless as well,

$$\operatorname{Tr} h^{+-} + \operatorname{Tr} h^{-+} = 0$$

Our variational argument does not depend neither on the trace of the fields nor the trace of the original hamiltonian. Then

$$\begin{aligned}
\hat{H}_{\mathrm{HF}} &= \sum_{\alpha,\beta}\left[h^{+-}_{\alpha\beta}\hat{c}^\dagger_\alpha \hat{c}_\beta + h^{-+}_{\alpha\beta}\hat{c}_\alpha \hat{c}^\dagger_\beta\right] + (\text{same sign terms}) \\
&= \sum_{\alpha,\beta}\left[h^{+-}_{\alpha\beta}\left(\delta_{\alpha\beta} - \hat{c}_\beta\hat{c}^\dagger_\alpha\right) + h^{-+}_{\alpha\beta}\left(\delta_{\alpha\beta} - \hat{c}^\dagger_\beta\hat{c}_\alpha\right)\right] + (\text{same sign terms}) \\
&= -\sum_{\alpha,\beta}\left[h^{+-}_{\beta\alpha}\hat{c}_\alpha\hat{c}^\dagger_\beta + h^{-+}_{\beta\alpha}\hat{c}^\dagger_\alpha\hat{c}_\beta\right] + (\text{same sign terms})
\end{aligned}$$

In second row we got rid of the traces sum and exchanged indices $\alpha \leftrightarrow \beta$. Confronting last and first row we conclude:

$$h^{+-}_{\beta\alpha} = -h^{-+}_{\alpha\beta} \qquad h^{-+}_{\beta\alpha} = -h^{+-}_{\alpha\beta} \tag{2.9}$$

thus, also on-diagonal fields are linked. We are allowed to determine, say, just $h^{+-}$: its counterpart $h^{-+}$ is completely determined. These properties were all we needed to get to the main argument of the method.

### 2.2.2 Self-consistent variational solution

We follow and extend the variational scheme proposed by Fabrizio [28]. Let:

$$\Delta^{pq}_{\alpha\beta} \equiv \left\langle \hat{a}^p_\alpha \hat{a}^q_\beta \right\rangle$$

thus, explicitly

$$\Delta^{+-}_{\alpha\beta} = \left\langle \hat{c}^\dagger_\alpha \hat{c}_\beta \right\rangle \qquad \Delta^{-+}_{\alpha\beta} = \left\langle \hat{c}_\alpha \hat{c}^\dagger_\beta \right\rangle \qquad \Delta^{++}_{\alpha\beta} = \left\langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \right\rangle \qquad \Delta^{--}_{\alpha\beta} = \left\langle \hat{c}_\alpha \hat{c}_\beta \right\rangle \tag{2.10}$$

The first two are "normal densities"; the second two are "anomalous densities". Conjugation of the last two gives a symmetry relation similar to Eq. (2.8):

$$\left(\Delta^{++}_{\alpha\beta}\right)^\dagger = \Delta^{--}_{\beta\alpha} \tag{2.11}$$

The HF ground-state energy is given by:

$$\begin{aligned} E_{\mathrm{HF}} &= \left\langle \hat{H}_{\mathrm{HF}} \right\rangle \\ &= \sum_{\alpha,\beta} \sum_{p,q} h^{pq}_{\alpha\beta} \Delta^{pq}_{\alpha\beta} \end{aligned}$$

Here $\Delta^{pq}_{\alpha\beta}$ are functions of the fields $h^{pq}_{\alpha\beta}$. Derive energy:

$$\begin{aligned} \frac{\partial E_{\mathrm{HF}}}{\partial h^{rs}_{\mu\nu}} &= \frac{\partial}{\partial h^{rs}_{\mu\nu}} \sum_{\alpha,\beta} \sum_{p,q} h^{pq}_{\alpha\beta} \Delta^{pq}_{\alpha\beta} \\ &= \Delta^{rs}_{\mu\nu} + \sum_{\alpha,\beta} \sum_{p,q} h^{pq}_{\alpha\beta} \frac{\partial \Delta^{pq}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} \end{aligned}$$

From Hellman-Feynman theorem [26], we know that for a parametric hamiltonian it holds:

$$\begin{aligned} \frac{\partial E_{\mathrm{HF}}}{\partial h^{rs}_{\mu\nu}} &= \langle \Psi | \frac{\partial \hat{H}_{\mathrm{HF}}}{\partial h^{rs}_{\mu\nu}} | \Psi \rangle \\ &= \langle \Psi | \frac{\partial}{\partial h^{rs}_{\mu\nu}} \sum_{\alpha,\beta} \sum_{p,q} h^{pq}_{\alpha\beta} \hat{a}^p_\alpha \hat{a}^q_\beta | \Psi \rangle \\ &= \left\langle \hat{a}^r_\mu \hat{a}^s_\nu \right\rangle \\ &= \Delta^{rs}_{\mu\nu} \end{aligned}$$

The last two results imply:

$$\sum_{\alpha,\beta} \sum_{p,q} h^{pq}_{\alpha\beta} \frac{\partial \Delta^{pq}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} = 0 \tag{2.12}$$

This must hold for any possible choice of fields $h^{pq}_{\alpha\beta}$. Note that we did not impose the energy derivative to be zero: it is not the HF energy we are minimising.

Let us get back to the actual hamiltonian $\hat{H}$. To reduce it to a quadratic form is equivalent to assume that its ground-state has the properties described in previous sections. Then the following decomposition must hold:

$$
\begin{aligned}
E &= \langle \hat{H} \rangle \\
&= \sum_{\alpha,\beta} A_{\alpha\beta} \langle \hat{c}^\dagger_\alpha \hat{c}_\beta \rangle + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta} \langle \hat{c}^\dagger_\alpha \hat{c}^\dagger_\beta \hat{c}_\gamma \hat{c}_\delta \rangle \\
&= \sum_{\alpha,\beta} A_{\alpha\beta} \langle \hat{a}^+_\alpha \hat{a}^-_\beta \rangle + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta} \left( \langle \hat{a}^+_\alpha \hat{a}^+_\beta \rangle \langle \hat{a}^-_\gamma \hat{a}^-_\delta \rangle - \langle \hat{a}^+_\alpha \hat{a}^-_\gamma \rangle \langle \hat{a}^+_\beta \hat{a}^-_\delta \rangle + \langle \hat{a}^+_\alpha \hat{a}^-_\delta \rangle \langle \hat{a}^+_\beta \hat{a}^-_\gamma \rangle \right) \\
&= \sum_{\alpha,\beta} A_{\alpha\beta} \Delta^{+-}_{\alpha\beta} + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta} \left( \Delta^{++}_{\alpha\beta} \Delta^{--}_{\gamma\delta} - \Delta^{+-}_{\alpha\gamma} \Delta^{+-}_{\beta\delta} + \Delta^{+-}_{\alpha\delta} \Delta^{+-}_{\beta\gamma} \right) \\
&= \sum_{\alpha,\beta} A_{\alpha\beta} \Delta^{+-}_{\alpha\beta} + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} \left( B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta} \right) \Delta^{+-}_{\alpha\beta} \Delta^{+-}_{\gamma\delta} + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta} \Delta^{++}_{\alpha\beta} \Delta^{--}_{\gamma\delta}
\end{aligned}
$$

In second passage we inserted the the relations (2.10). Take the derivative:

$$
\begin{aligned}
\frac{\partial E}{\partial h^{rs}_{\mu\nu}} = \sum_{\alpha,\beta} A_{\alpha\beta} \frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} + \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} \left( B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta} \right) \left( \frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} \Delta^{+-}_{\gamma\delta} + \Delta^{+-}_{\alpha\beta} \frac{\partial \Delta^{+-}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}} \right) \\
+ \frac{1}{2} \sum_{\alpha,\beta,\gamma,\delta} B_{\alpha\beta\gamma\delta} \left( \frac{\partial \Delta^{++}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} \Delta^{--}_{\gamma\delta} + \Delta^{++}_{\alpha\beta} \frac{\partial \Delta^{--}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}} \right)
\end{aligned}
$$

Using Eq. (2.5) we end up with:

$$
\begin{aligned}
\frac{\partial E}{\partial h^{rs}_{\mu\nu}} = \sum_{\alpha,\beta} \left[ A_{\alpha\beta} + \sum_{\gamma,\delta} \left( B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta} \right) \Delta^{+-}_{\gamma\delta} \right] \frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} \\
+ \sum_{\alpha,\beta,\gamma,\delta} \frac{B_{\alpha\beta\gamma\delta} \Delta^{--}_{\gamma\delta}}{2} \frac{\partial \Delta^{++}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} + \sum_{\alpha,\beta,\gamma,\delta} \frac{B_{\alpha\beta\gamma\delta} \Delta^{++}_{\alpha\beta}}{2} \frac{\partial \Delta^{--}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}}
\end{aligned}
$$

Let now:

$$
C_{\alpha\beta} \equiv \frac{1}{2} \left[ A_{\alpha\beta} + \sum_{\gamma,\delta} \left( B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta} \right) \Delta^{+-}_{\gamma\delta} \right] \qquad D_{\alpha\beta} \equiv \sum_{\gamma,\delta} \frac{B_{\alpha\beta\gamma\delta} \Delta^{--}_{\gamma\delta}}{2} \tag{2.13}
$$

From Fermi anticommutation rules, we have:

$$
\Delta^{+-}_{\alpha\beta} = \delta_{\alpha\beta} - \Delta^{-+}_{\beta\alpha} \quad \Longrightarrow \quad \frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} = - \frac{\partial \Delta^{-+}_{\beta\alpha}}{\partial h^{rs}_{\mu\nu}}
$$

which implies:

$$
\begin{aligned}
\sum_{\alpha,\beta} 2C_{\alpha\beta} \frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} &= \sum_{\alpha,\beta} \left( C_{\alpha\beta} \frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} + C_{\alpha\beta} \frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} \right) \\
&= \sum_{\alpha,\beta} \left( C_{\alpha\beta} \frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} - C_{\beta\alpha} \frac{\partial \Delta^{-+}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} \right)
\end{aligned}
$$

Moreover, using Eqns. (2.4), (2.11),

$$
\sum_{\alpha,\beta} \frac{B_{\alpha\beta\gamma\delta} \Delta^{++}_{\alpha\beta}}{2} = \sum_{\alpha,\beta} \left( \frac{B_{\delta\gamma\beta\alpha} \Delta^{--}_{\beta\alpha}}{2} \right)^*
$$

Using these two relations, we conclude:

$$
\sum_{\alpha,\beta,\gamma,\delta} \frac{B_{\alpha\beta\gamma\delta} \Delta^{++}_{\alpha\beta}}{2} \frac{\partial \Delta^{--}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}} = \sum_{\gamma,\delta} D^*_{\delta\gamma} \frac{\partial \Delta^{--}_{\gamma\delta}}{\partial h^{rs}_{\mu\nu}}
$$

Recollect everything and impose the solution to be a stationary point of energy,

$$\frac{\partial E}{\partial h^{rs}_{\mu\nu}} \stackrel{!}{=} 0$$

The notation "$\stackrel{!}{=}$" indicates, throughout the thesis, that we are imposing some condition. In the end we have:

$$\frac{\partial E}{\partial h^{rs}_{\mu\nu}} = \sum_{\alpha,\beta} \left( C_{\alpha\beta} \frac{\partial \Delta^{+-}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} - C_{\beta\alpha} \frac{\partial \Delta^{-+}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} + D_{\alpha\beta} \frac{\partial \Delta^{++}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} + D^{*}_{\beta\alpha} \frac{\partial \Delta^{--}_{\alpha\beta}}{\partial h^{rs}_{\mu\nu}} \right) \stackrel{!}{=} 0 \tag{2.14}$$

Now compare Eqns. (2.12) and (2.14). We recognise $h^{+-}_{\alpha\beta} = C_{\alpha\beta}$ and $h^{++}_{\alpha\beta} = D_{\alpha\beta}$; the remaining fields are determined by the symmetry relations (2.9) and (2.8). Explicitly, the fields are self-consistently determined by the following two equations:

$$h^{+-}_{\alpha\beta} = \frac{1}{2} \left[ A_{\alpha\beta} + \sum_{\gamma,\delta} \left( B_{\alpha\gamma\delta\beta} - B_{\alpha\gamma\beta\delta} \right) \Delta^{+-}_{\gamma\delta} \right] \tag{2.15}$$

$$h^{++}_{\alpha\beta} = \sum_{\gamma,\delta} \frac{B_{\alpha\beta\gamma\delta} \Delta^{--}_{\gamma\delta}}{2} \tag{2.16}$$

The right-hand side of both equations depends on the fields themselves: $\Delta^{pq}_{\alpha\beta}$ are defined as expectation values of two-points fermionic operators over the ground-state $|\Psi\rangle$. For a given set of fields $h^{pq}_{\alpha\beta}$, the latter is determined (up to some unitary transformation) as the ground-state of a fermionic non-interacting model. In other words, being $\mathbf{h}$ the tensor containing all fields,

$$\Delta^{pq}_{\alpha\beta}[\mathbf{h}] = \langle \Psi[\mathbf{h}] | \hat{a}^{p}_{\alpha} \hat{a}^{q}_{\beta} | \Psi[\mathbf{h}] \rangle \tag{2.17}$$

This set of equations defines our computational problem. We are searching a set of parameters $\{\Delta^{pq}_{\alpha\beta}\}$ such that, determining the fields via Eqns. (2.15), (2.16), then Eq. (2.17) is respected. **The great advantage of this scheme is that, with an appropriate choice of the Hilbert space basis (for example, moving to reciprocal space for a translationally-invariant model), we can implement easily model symmetries.**

The list of parameters $\{\Delta^{pq}_{\alpha\beta}\}$ to be determined self-consistently may be not-so-short: by looking at Eq. (2.17), the indices $\alpha, \beta, p, q$ are spanning a large Hilbert space. However, this equation can be manipulated and transformed, reducing the set of parameters to be determined self-consistently to a very manageable number. For each phase we define these last as "Hartree-Fock parameters" (HFPs): a set of objects self-consistently determined by a set equivalent, but transformed, version of Eq. (2.17).

## 2.3 Sketch of the iHF algorithm

In this section we give a practical sketch of the algorithm we use to compute Eq. (2.17) for all given phases. We refer to this technique as "iterative Hartree Fock" (iHF). Let $\mathbf{v}$ be the vector containing all HFPs. Suppose we have $N$ HFPs $v_i$: then

$$\mathbf{v} \in \mathbb{C}^N \qquad \mathbf{v} \equiv \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_N \end{bmatrix}$$

In order to get a physical solution, each HFP must satisfy a self-consistency equation similar to Eq. (2.17). Each equation has the structure:

$$v_i = (\text{Some non-linear function of } \mathbf{v})$$

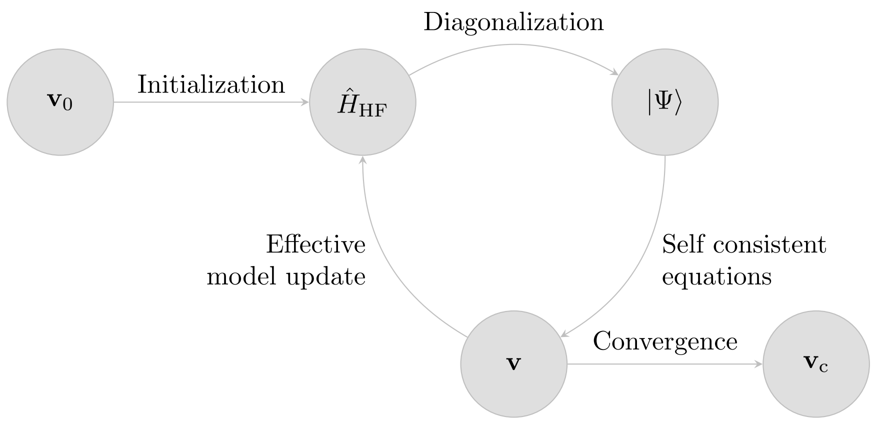



Figure 2.1 Schematisation of the self-consistent Hartree Fock method described in Sec. 2.3. The effective model is solved, the solution is used to estimate self-consistently the Hartree Fock vector $\mathbf{v}$, which is then used to update iteratively the model itself. The notation $\mathbf{v}_0$ indicates the initialiser HFPs vector, $\mathbf{v}_c$ the converged HFPs vector.

Let A be the non-linear operator collecting all these functions. The problem of finding a self-consistent solution can be reformulated as follows: $\mathbf{v}$ represents a physical solution if it is a fixed point of the map A,

$$\mathrm{A}\colon \mathbb{C}^N \to \mathbb{C}^N \qquad \mathrm{A}(\mathbf{v}) = \mathbf{v} \qquad \Longrightarrow \qquad \mathbf{v} \text{ is a solution.} \tag{2.18}$$

We implement a rather easy scheme: first, we obtain an analytic expression for A. Then we choose an initialiser $\mathbf{v}_0$ and apply A to it, obtaining a new HFPs vector. Iterating this procedure, we "follow" the non-linear map A until we claim to have converged. Let us explain in detail:

1. Start with a generic quartic hamiltonian:
$$\hat{H} = \sum_{\alpha\beta} A_{\alpha\beta}\hat{c}^\dagger_\alpha \hat{c}_\beta + \sum_{\alpha\beta\gamma\delta} B_{\alpha\beta\gamma\delta}\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta\hat{c}_\gamma\hat{c}_\delta$$
Choose the model parameters and set an electronic density;

2. Following the discussion of Sec. 2.2, approximate analytically $\hat{H}$ with a quadratic hamiltonian
$$\hat{H} \simeq \hat{H}_{\mathrm{HF}} \equiv \sum_{\alpha\beta}\left[h^{+-}_{\alpha\beta}\hat{c}^\dagger_\alpha\hat{c}_\beta - h^{+-}_{\beta\alpha}\hat{c}_\alpha\hat{c}^\dagger_\beta + h^{++}_{\alpha\beta}\hat{c}^\dagger_\alpha\hat{c}^\dagger_\beta + \left(h^{++}_{\beta\alpha}\right)^*\hat{c}_\alpha\hat{c}_\beta\right]$$
with the fields given by Eqns. (2.15), (2.16);

3. Impose selection rules on the fields $h^{+-}_{\alpha\beta}$, $h^{++}_{\alpha\beta}$, eventually eliminating those not respecting the symmetry sector we are working in. As an example: if we choose not to break total charge conservation, set $\forall\alpha,\beta\colon h^{++}_{\alpha\beta} = 0$;

4. Choose a starting value for the HFPs vector $\mathbf{v}_0$;

5. Set $\mathbf{v} = \mathbf{v}_0$ and enter the iterative scheme of Fig. 2.1:
   a) Get a numeric approximation of the hamiltonian for the current value of $\mathbf{v}$, $\hat{H}_{\mathrm{HF}}[\mathbf{v}]$;
   b) Determine the value $\mu[\mathbf{v}]$ giving the desired density;
   c) Diagonalise the effective hamiltonian $\hat{H}_{\mathrm{HF}}[\mathbf{v}] - \mu[\mathbf{v}]\hat{N}$, and obtain the approximated ground-state $|\Psi[\mathbf{v}]\rangle$;
   d) Compute $\mathrm{A}(\mathbf{v})$ using self-consistency equations;
   e) If the algorithm converged, exit; otherwise, update the value of $\mathbf{v}$ repeat from point a).

We voluntarily left unspecified three details:

- How is the chemical potential $\mu$ determined?
- What criterion do we use to claim that the algorithm "converged"?
- How is the HFPs vector $\mathbf{v}$ updated?

The sections that follows deal with these questions.

### 2.3.1 Fixed density simulations and the role of $\mu$

In this thesis all derivations and simulations are carried out at fixed density. This requires a way to fix $\mu$ during iterations of the algorithm. This must be done iteratively: changing at each step the HFPs vector, we are actively modifying the approximate hamiltonian. Chemical potential must be determined accordingly to maintain the required density.

For a given choice of HFPs vector and chemical potential, the ground state $|\Psi[\mathbf{v},\mu]\rangle$ is uniquely determined; then we compute density as

$$n(\mathbf{v},\mu) \equiv \frac{1}{\mathcal{V}} \sum_{\alpha} \langle \Psi[\mathbf{v},\mu] | \hat{c}^\dagger_\alpha \hat{c}_\alpha | \Psi[\mathbf{v},\mu] \rangle$$

being $\mathcal{V}$ the volume. Suppose we fix density to a value $n_0 \in [0,1]$. At step $i$, naming the current HFPs vector $\mathbf{v}_i$, $\mu_i$ (the chemical potential at current step) is determined as

$$\mu_i \equiv \min_{\mu} |n(\mathbf{v}_i,\mu) - n_0| \tag{2.19}$$

The density $n$ must be a monotonically increasing function of $\mu$: then we can translate the above equation, which would inspire a minimisation procedure, to a much simpler node-search rule:

$$\mu_i = \mu \mid_{\delta n(\mathbf{v}_i,\mu)=0} \quad \text{being} \quad \delta n(\mathbf{v}_i,\mu) \equiv n(\mathbf{v}_i,\mu) - n_0$$

In order to where the function changes sign (which is, the position of $\mu_i$) we use a basic bisection algorithm:

1. Set an arbitrary tolerance $\Delta n > 0$ and a shifting parameter $\Delta\mu > 0$;
2. Choose starting lower and upper bounds $\mu_\mathrm{L} < \mu_\mathrm{U}$;
3. Compute $\delta n(\mathbf{v}_i,\mu_\mathrm{L})$. Evaluate:
   - if $|\delta n(\mathbf{v}_i,\mu_\mathrm{L})| \leq \Delta n$, set $\mu_i = \mu_\mathrm{L}$ and halt;
   - if $\delta n(\mathbf{v}_i,\mu_\mathrm{L}) < -\Delta n$ do nothing;
   - if $\delta n(\mathbf{v}_i,\mu_\mathrm{L}) > \Delta n$, shift the lower bound down, $\mu_\mathrm{L} \to \mu_\mathrm{L} - \Delta\mu$ and repeat from step 3;
4. Compute $\delta n(\mathbf{v}_i,\mu_\mathrm{U})$. Evaluate:
   - if $|\delta n(\mathbf{v}_i,\mu_\mathrm{U})| \leq \Delta n$, set $\mu_i = \mu_\mathrm{U}$ and halt;
   - if $\delta n(\mathbf{v}_i,\mu_\mathrm{U}) > \Delta n$ do nothing;
   - if $\delta n(\mathbf{v}_i,\mu_\mathrm{U}) < -\Delta n$, shift the upper bound up, $\mu_\mathrm{U} \to \mu_\mathrm{U} + \Delta\mu$ and repeat from step 4;
5. At this point, the the lower bound underestimates density while the upper bound overestimates. Start iterations:
   a) Compute the midpoint,
   $$\mu_\mathrm{M} = \frac{\mu_\mathrm{L} + \mu_\mathrm{U}}{2}$$
   b) Evaluate:
      - If $|\delta n(\mathbf{v}_i,\mu_\mathrm{M})| \leq \Delta n$, set $\mu_i = \mu_\mathrm{M}$ and halt;
      - If $\delta n(\mathbf{v}_i,\mu_\mathrm{M}) > \Delta n$, set $\mu_\mathrm{U} = \mu_\mathrm{M}$ and repeat from point a);
      - If $\delta n(\mathbf{v}_i,\mu_\mathrm{M}) < -\Delta n$, set $\mu_\mathrm{L} = \mu_\mathrm{M}$ and repeat from point a);

The bottleneck of the procedure is our ability to compute expectation values. This depends on how much we are able to simplify analytically the computations for each phase.

As will turn out in the derivation, chemical potential will be renormalised by interaction. This means that applying self-consistent HF methods to the EHM will produce an effective hamiltonian containing a shift to the chemical potential,

$$\mu \to \mu + F[\mathbf{v}] \equiv \tilde{\mu}[\mathbf{v}]$$

being $F$ some function of the HFPs. The chemical potential $\mu$ is an example of Renormalised Model Parameter (RMP). As we will see, also the hopping $t$ will turn out to be renormalised by interactions.

Note that at step $i$ of the iHF algorithm, our bisection procedure determines $\tilde{\mu}_i$, not $\mu_i$. This consideration is relevant when dealing with ground state energy estimation. Let us abstract from the EHM and iHF, and consider a generic hamiltonian $\hat{H}$. To find the ground state $|\Psi\rangle$ means precisely to solve

$$\frac{\delta}{\delta \langle\Psi|} \langle\Psi|\hat{H}|\Psi\rangle = 0$$

Now, working with a fixed particles number $N$ means we are performing a constrained minimisation (which is, taking the functional derivative) on the composite operator

$$\hat{H} - \mu(\hat{N} - N) \equiv \hat{K} + \mu N$$

being $\mu$ the Lagrange multiplier, $\hat{N}$ the number operator and $\hat{K} \equiv \hat{H} - \mu\hat{N}$. Finding the constrained ground state parametrised by a vector $\mathbf{v}$ now means to solve

$$\mathbf{v} \implies |\Psi[\mathbf{v}]\rangle \quad : \quad \begin{cases} \left.\dfrac{\delta}{\delta \langle\Psi|} \langle\Psi|\hat{K} + \mu N|\Psi\rangle \right|_{|\Psi\rangle = |\Psi[\mathbf{v}]\rangle} = 0 \\ \langle\Psi[\mathbf{v}]|\hat{N}|\Psi[\mathbf{v}]\rangle = N \end{cases}$$

In the first line $\mu N$ is derived to zero; the constraint is imposed in the second line. Energy is given by:

$$\begin{aligned} \langle\hat{H}\rangle &= \langle\Psi[\mathbf{v}]|\hat{K}|\Psi[\mathbf{v}]\rangle + \mu N \\ &= \Omega[\mathbf{v}] + (\tilde{\mu}[\mathbf{v}] - F[\mathbf{v}])N \end{aligned}$$

being $\Omega$ the expectation value (EV) of $\hat{K}$. To correctly compute energy, we need to add $+\mu N$; since we determine $\tilde{\mu}[\mathbf{v}] \neq \mu$, everything needs to be expressed in terms of it. Using this relation, when the algorithm has converged, we can estimate ground-state energy.

### 2.3.2 The reject-mix-iterate convergence scheme

Let us discuss briefly the convergence criterion of our algorithm and the iteration mechanism. The iHF scheme we design is a sequential loop that runs up to a logical halt condition. In order to specify such condition, we need to define our notion of "convergence". Recall the definition of $\mathrm{A}(\mathbf{v})$, the non-linear map in HFPs space whose component are given by a set of self-consistency equations: we claim to have reached convergence when $\mathbf{v}$ comes "close enough" to a fixed point of the map A, as in Eq. (2.18). Explicitly, we stop whenever the following logic condition is satisfied for each component of $\mathbf{v} = (v_1, v_2, \cdots)$:

$$\forall v_j \quad : \quad |v_j - \mathrm{A}_j(\mathbf{v})| \leq \Delta v_j$$

being $\mathrm{A}(\mathbf{v})$ the map image of the HFP vector $\mathbf{v}$, and $\mathrm{A}_j$ its $j$-th component. The tolerance $\Delta v_j$ is arbitrary. If just one component of $\mathbf{v}$ is mapped "too far" (with respect to $\Delta v_j$) the step is rejected and iterations continue. In this text we deal in with quantities of order $10^{-1}$, so convergence values of order $10^{-3} \div 10^{-4}$ will be in general sufficient.

At step $i$, we check the distance of the two vectors $\mathrm{A}(\mathbf{v}_i)$ and $\mathbf{v}_i$ and if possible, halt. If instead convergence is not reached, we need repeat the cycle with a new initialiser $\mathbf{v}_{i+1}$. A very simple choice would be to set $\mathbf{v}_{i+1} = \mathbf{v}_i$. We do something slightly different:

$$\mathbf{v}_{i+1} \equiv g\mathrm{A}(\mathbf{v}_i) + (1-g)\mathbf{v}_i \tag{2.20}$$

The "mixing parameter" $g$ is a simple but useful knob to tune algorithmic evolution. Its usefulness comes from this observation: even if a fixed point exists, we don't know how easy it is to actually get there following A. Setting $g = 1$, the new initialiser is just the image of the previous initialiser. By moving "slower" ($g < 1$) with an accurate optimisation of the mixing, it is possible to reach convergence in a controlled way.

Let us give some concrete justification about the relevance of controlling $g$. When it comes to HF iterative schemes of this kind, it can sometimes happen for the algorithm to remain stuck in infinite loops. An example of this behaviour is given in Sec. **??**. A typical situation is the following: during algorithm evolution we reach a couple of points $\mathbf{v}^{(a)}$, $\mathbf{v}^{(b)}$ such that

$$\ldots \xrightarrow{\mathrm{A},g} \mathbf{v}^{(a)} \xrightarrow{\mathrm{A},g} \mathbf{v}^{(b)} \xrightarrow{\mathrm{A},g} \mathbf{v}^{(a)} \xrightarrow{\mathrm{A},g} \mathbf{v}^{(b)} \xrightarrow{\mathrm{A},g} \ldots$$

With the notation "$\xrightarrow{\mathrm{A},g}$" we intend one algorithmic step at mixing parameter $g$. We are stuck forever in this loop. Now, to predict if couples like $(\mathbf{v}^{(a)}, \mathbf{v}^{(b)})$ exist, how they depend on $g$, where they are located in HFPs space... is impossible at this level. Thus there is no immediate way to avoid these loops *a priori*. One workaround we found, however, was to just make $g$ smaller. We gained empirical evidence that to "slow down" in HFPs space makes numeric instabilities less frequent. The trade-off, however, is that if we move too slow, runtime could get very large. Note that inaccurate setup of the algorithm could lead to resources-intensive runs also in "ordinary" situations, when we are not stuck in a loop.

To keep the number of iterations under control, we also defined an upper bound $p$ to the total number of steps. If after $p$ steps we algorithm failed to converge, it halts. In that case, $\mathbf{v}$ is saved as a "`NaN`" (Not a number) vector and we move to another simulation. Introducing this simple rule, runtime is controlled and we gain an easy method to identify the regions of parameter space requiring more care – just read from the raw data where `NaN`s have been saved and repeat simulations fine-tuning the parameters. This basically means to lower $g$ and increase $p$. These considerations end this technical introductory chapter on iHF methods.

# Chapter 3

# Phase transitions and SSB in EHM

The general idea of this project is to apply HF methods to the EHM, derive analytically the approximated quadratic hamiltonian of Eq. (2.6) and then apply the iHF procedure sketched in Sec. 2.3 to extract the self-consistent values of the various HFPs. One of the advantages of using iHF methods is that we can impose rather simply the symmetries of the HF approximated wavefunction.

This is most useful when dealing with competing phases. A phase transition is basically described by some spontaneous symmetry breaking (SSB) and the rising to a non-zero value of an order parameter. We will deal with three phases:

1. the normal phase, where no symmetry is broken;
2. the antiferromagnetic phase, where we allow for spin density to be non-uniform across sites;
3. the superconducting phase, where we do not conserve particle number.

Each phase is discussed in a separate chapter. Many other "phases" are possible, based on which symmetries are respected and which are not.

For each phase, we will extract the best approximation to the true ground state wavefunction. In computational terms, we will find three solutions to the general self-consistency problem of Eq. (2.17). In the end we want to compare the free energy of these three solutions, assessing which one is the most stable. Any HF approximation, being by construction not the true ground state, overestimates energy. Thus, comparing energies of our solutions we get a hint of the features of the true ground state.

## 3.1 The EHM symmetries

In this section we present the symmetries of the extended Hubbard model introduced in Eq. (1.2)

$$\hat{H}_{\mathrm{EHM}} = \underbrace{-t\sum_{\langle ij\rangle}\sum_{\sigma}\hat{c}^{\dagger}_{i\sigma}\hat{c}_{j\sigma}}_{\hat{H}_t} \underbrace{+U\sum_{i}\hat{n}_{i\uparrow}\hat{n}_{i\downarrow}}_{\hat{H}_U} \underbrace{-V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{n}_{i\sigma}\hat{n}_{j\sigma'}}_{\hat{H}_V}$$

The symmetries of the model can be divided in two groups: the ones related to the geometry of the square lattice, and those related to the algebraic properties of $\hat{H}_{\mathrm{EHM}}$. We start by the lattice symmetries.

### 3.1.1 Lattice symmetries

We study the EHM on the planar square lattice, with one orbital per site. We indicate with $L_x$ the number of sites in direction $x$, and $L_y$ in direction $y$. The hamiltonian of Eq. (1.2) respects three fundamental lattice symmetries:

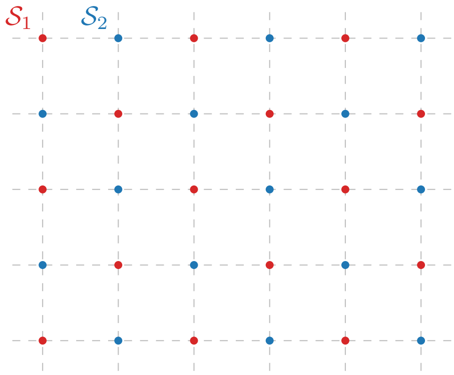



Figure 3.1 | Square lattice $\mathcal{L}$ bipartition in two sublattices $\mathcal{S}_1$ (red), $\mathcal{S}_2$ (blue).

1. Discrete translations symmetry in both the $x$ and $y$ directions. We refer to the groups of one-site translations as as $\mathrm{T}_1^{(x)}$, $\mathrm{T}_1^{(x)}$. It holds:

$$\mathrm{T}_n^{(\ell)} \supset \mathrm{T}_{kn}^{(\ell)} \qquad k \in \mathbb{N} \tag{3.1}$$

A shift in direction $\ell$ by $kn$ sites can be decomposed in $k$ shift by $n$ sites. Invariance by $n$-sites shifts, thus, implies invariance also for $kn$-sites shifts.

2. Discrete $\pi/2$ rotations symmetry. This symmetry is described by the four-fold cyclic point group $\mathrm{C}_4$, containing rotations by angle $\pi/2, \pi, 3\pi/2$ and identity. A similar property as for the translations holds for cyclic point groups,

$$\mathrm{C}_n \supset \mathrm{C}_{n/k} \qquad k, n/k \in \mathbb{N} \tag{3.2}$$

Invariance for rotations of angle $2\pi/n$ implies invariance for rotations of angle $k \times 2\pi/n$.

3. Inversion symmetry. We define $\mathrm{I}_2$ as the inversion group in two-dimensional space, containing the reflection operator and identity. This point may appear redundant, because

$$\mathrm{I}_2 \equiv \mathrm{C}_2 \tag{3.3}$$

In 2D space, to rotate by $\pi$ around the origin and to mirror the position of a point are the same operation.

The planar square lattice is bipartite. Let us give a definition. Consider a graph $\mathcal{L}$ and the set of its vertices $V(\mathcal{L}) = \{v_i\}$. Let $E(\mathcal{L})$ be the set of graph edges,

$$E(\mathcal{L}) = \{(v, w) \mid v, w \in V(\mathcal{L}) \wedge v,\ w \text{ are connected}\}$$

The graph is said to be bipartite if two sub-graphs $\mathcal{S}_1$, $\mathcal{S}_2$ exist such that:

- The vertices of $\mathcal{L}$ split in the two sub-graphs,

$$V(\mathcal{L}) = V(\mathcal{S}_1) \cup V(\mathcal{S}_2)$$

- Vertices of one sub-graph are linked only to vertices of the other sub-graph. In other words

$$v, w \in \mathcal{S}_i \quad \Longrightarrow \quad (v, w) \notin E(\mathcal{L})$$

The planar square lattice is bipartite: it can be divided in two square lattices, as represented in Fig. 3.1. Two sites are "connected" if linked by a non-zero hopping operator. Which is, if they are NNs. Note that one-site translations leave the hamiltonian invariant but exchange the two sublattices, while two-sites translations don't.

### 3.1.2 Spin and charge symmetries

The conventional HM shows a global U(2) symmetry, extended to $\mathrm{SU}(2)\times\mathrm{SU}(2)/\mathbb{Z}_2 = \mathrm{SO}(4)$ symmetry at half-filling on a bipartite lattice [1], such as the square lattice. For the EHM, we limit to prove that we can identify a $\mathrm{U}(2)\subset\mathrm{SO}(4)$ global symmetry.

**Gauge symmetry.** The EHM is symmetric under the global transformation

$$\hat{c}_{i\sigma}\to e^{-i\eta}\hat{c}_{i\sigma}\qquad \hat{c}^\dagger_{i\sigma}\to e^{i\eta}\hat{c}^\dagger_{i\sigma}\qquad \eta\in\mathbb{R} \tag{3.4}$$

To prove this, it suffices to note that

$$\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma'}\quad\to\quad e^{i\eta}\hat{c}^\dagger_{i\sigma}e^{-i\eta}\hat{c}_{j\sigma'}=\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma'} \tag{3.5}$$

As a consequence, any product of fermionic operators with an equal number of annihilations and creations is symmetric. Hence the kinetic part $\hat{H}_t$ and the interactions $\hat{H}_U$, $\hat{H}_V$ are invariant. As a consequence, the total number of particles (charge)

$$\hat{N}=\sum_{i\sigma}\hat{c}^\dagger_{i\sigma}\hat{c}_{i\sigma}$$

is a good quantum number. The symmetry is described by the transformation of Eq. (3.4), which is a pure phase shift by $\eta$. The relative group is $\mathrm{U}^{\mathrm{c}}(1)$, with a superscript "c" indicating "charge".

**Global spin symmetry** The EHM is described by three operators: NN hopping, independent of $\sigma$; local repulsion between opposite-spin densities; non-local attraction, independent of $\sigma$. If we rotate all spins by an equal amount, the hamiltonian must remain identical. An identical observation holds for the number operator, $\hat{N}$. Thus $\hat{H}_{\mathrm{EHM}}-\mu\hat{N}$ must be exhibit a global SU(2) symmetry, related to the rotation in 3D space of all spins altogether.

Let us prove this rigorously. Let the on-site spinor:

$$\hat{\Psi}_i\equiv\begin{bmatrix}\hat{c}_{i\uparrow}\\ \hat{c}_{i\downarrow}\end{bmatrix}$$

and the Pauli matrices

$$\tau^x\equiv\begin{bmatrix} & 1\\ 1 & \end{bmatrix}\qquad \tau^y\equiv\begin{bmatrix} & -i\\ i & \end{bmatrix}\qquad \tau^z\equiv\begin{bmatrix}1 & \\ & -1\end{bmatrix}$$

We define the local spin operator as:

$$\hat{S}^\alpha_i\equiv\frac{1}{2}\hat{\Psi}^\dagger_i\tau^\alpha\hat{\Psi}_i\qquad \left[\hat{S}^\alpha_j,\hat{S}^\beta_j\right]=i\sum_\gamma\epsilon_{\alpha\beta\gamma}\hat{S}^\gamma_j$$

with $\epsilon_{\alpha\beta\gamma}$ the Levi-Civita tensor. It can be checked by explicit calculation that the $\mathfrak{su}(2)$ spin algebra is actually respected. For example, consider $\alpha=x$, $\beta=y$:

$$\begin{aligned}
\left[\hat{S}^x_j,\hat{S}^y_j\right] &= \frac{1}{4}\left[\hat{\Psi}^\dagger_j\begin{bmatrix} & 1\\ 1 & \end{bmatrix}\hat{\Psi}_j,\hat{\Psi}^\dagger_j\begin{bmatrix} & -i\\ i & \end{bmatrix}\hat{\Psi}_j\right]\\
&= \frac{1}{4}\left[\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow}+\hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow},-i\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow}+i\hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow}\right]\\
&= \frac{i}{4}\left[\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow},\hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow}\right]-\frac{i}{4}\left[\hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow},\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow}\right]\\
&= \frac{i}{2}\left[\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow},\hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow}\right]\\
&= \frac{i}{2}\left(\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow}\hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow}-\hat{c}^\dagger_{j\downarrow}\hat{c}_{j\uparrow}\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\downarrow}\right)\\
&= \frac{i}{2}\left(-\hat{c}^\dagger_{j\uparrow}\hat{c}^\dagger_{j\downarrow}\hat{c}_{j\downarrow}\hat{c}_{j\uparrow}+\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\uparrow}+\hat{c}^\dagger_{j\downarrow}\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\uparrow}\hat{c}_{j\downarrow}-\hat{c}^\dagger_{j\downarrow}\hat{c}_{j\downarrow}\right)\\
&= \frac{i}{2}\left(\hat{c}^\dagger_{j\uparrow}\hat{c}_{j\uparrow}-\hat{c}^\dagger_{j\downarrow}\hat{c}_{j\downarrow}\right)\\
&= i\hat{S}^z_j
\end{aligned}$$

We used Fermi anticommutation rules, $\{\hat{c}_{i\sigma}, \hat{c}^\dagger_{j\sigma'}\} = \delta_{ij}\delta_{\sigma\sigma'}$. Identical derivations can be carried out for all other components.

Let the raising and lowering operators:

$$\hat{S}^+_j \equiv \frac{\hat{S}^x_j + i\hat{S}^y_j}{2} \qquad \hat{S}^-_j \equiv \frac{\hat{S}^x_j - i\hat{S}^y_j}{2}$$

and the global spin operators

$$\hat{S}^\alpha \equiv \sum_{i\in\mathcal{L}} \hat{S}^\alpha_i \qquad \left[\hat{S}^\alpha, \hat{S}^\beta\right] = i\sum_\gamma \epsilon_{\alpha\beta\gamma}\hat{S}^\gamma$$

being $\mathcal{L}$ the lattice. Explicitly,

$$\hat{S}^z \equiv \frac{1}{2}\sum_{i\in\mathcal{L}} (\hat{n}_{i\uparrow} - \hat{n}_{i\downarrow}) \qquad \hat{S}^+ \equiv \frac{1}{2}\sum_{i\in\mathcal{L}} \hat{c}^\dagger_{i\uparrow}\hat{c}_{i\downarrow} \qquad \hat{S}^- \equiv \left(\hat{S}^+\right)^\dagger$$

The EHM hamiltonian and the number operator commutate with the global spin operator. For the kinetic part $\hat{H}_t$ and the local repulsion $\hat{H}_U$ this is a well known result [1, 2],

$$\left[\hat{H}_t + \hat{H}_U, \hat{\mathbf{S}}\right] = 0 \qquad \left[\hat{N}, \hat{\mathbf{S}}\right] = 0$$

For the non-local attraction, since it depends solely on total densities on neighbouring sites, spin-rotational invariance must hold. Anyway, let us give a short formal proof following this chain of results:

1. The non-local attraction can be written as:

$$\begin{aligned}\hat{H}_V &= V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'} \hat{n}_{i\sigma}\hat{n}_{j\sigma'} \\ &= V\sum_{\langle ij\rangle} \hat{\Psi}^\dagger_i\hat{\Psi}_i\hat{\Psi}^\dagger_j\hat{\Psi}_j \end{aligned} \tag{3.6}$$

   having we used $\hat{\Psi}^\dagger_i\hat{\Psi}_i = \hat{c}^\dagger_{i\uparrow}\hat{c}_{i\uparrow} + \hat{c}^\dagger_{i\downarrow}\hat{c}_{i\downarrow}$.

2. Similarly, the number operator is given by:

$$\begin{aligned}\hat{N} &= \sum_{i\in\mathcal{L}}\sum_{\sigma} \hat{n}_{i\sigma} \\ &= \sum_{i\in\mathcal{L}} \hat{\Psi}^\dagger_i\hat{\Psi}_i \end{aligned} \tag{3.7}$$

3. Let now $\hat{R}(\boldsymbol{\theta})$ be any global spin rotation,

$$\hat{R}(\boldsymbol{\theta}) \equiv \exp\left(-i\boldsymbol{\theta}\cdot\hat{\mathbf{S}}\right) \quad \text{where} \quad \boldsymbol{\theta} = (\theta_x, \theta_y, \theta_z)$$

   Spinors transform as:

$$\hat{R}(\boldsymbol{\theta})\hat{\Psi}_i\hat{R}(\boldsymbol{\theta})^\dagger = \mathcal{R}(\boldsymbol{\theta})\hat{\Psi}_i$$

   with $\mathcal{R}(\boldsymbol{\theta}) \in \mathrm{SU}(2)$ the unitary matrix associated to said rotation,

$$\mathcal{R}(\boldsymbol{\theta}) = \exp\left(-\frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\tau}\right) \quad \text{where} \quad \boldsymbol{\tau} = (\tau_x, \tau_y, \tau_z) \tag{3.8}$$

   Let us prove this. Let $\hat{A} = i\boldsymbol{\theta}\cdot\hat{\mathbf{S}}$ e $\hat{B} = \hat{\Psi}_i$. The following identity, a consequence of Baker-Campbell-Housdorff formula, holds:

$$e^{\hat{A}}\hat{B}e^{-\hat{A}} = \hat{B} + \left[\hat{A},\hat{B}\right] + \frac{1}{2!}\left[\hat{A},\left[\hat{A},\hat{B}\right]\right] + \frac{1}{3!}\left[\hat{A},\left[\hat{A},\left[\hat{A},\hat{B}\right]\right]\right] + \cdots$$

The first order of the expansion gives:

$$
\begin{aligned}
[i\boldsymbol{\theta}\cdot\hat{\mathbf{S}},\hat{\Psi}_i] &= i\sum_\alpha \theta_\alpha[\hat{S}^\alpha,\hat{\Psi}_i]\\
&= -\frac{i}{2}\sum_\alpha \theta_\alpha\tau^\alpha\hat{\Psi}_i\\
&= -\frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\tau}\hat{\Psi}_i
\end{aligned}
$$

We omitted some purely algebraic passages. The second order:

$$
\begin{aligned}
\Big[i\boldsymbol{\theta}\cdot\hat{\mathbf{S}},\Big[i\boldsymbol{\theta}\cdot\hat{\mathbf{S}},\hat{\Psi}_i\Big]\Big] &= \Big[i\boldsymbol{\theta}\cdot\hat{\mathbf{S}},-\frac{i}{2}(\boldsymbol{\theta}\cdot\boldsymbol{\tau})\hat{\Psi}_i\Big]\\
&= -\frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\tau}\Big[i\boldsymbol{\theta}\cdot\hat{\mathbf{S}},\Psi_i\Big]\\
&= \frac{1}{4}(\boldsymbol{\theta}\cdot\boldsymbol{\tau})^2\hat{\Psi}_i
\end{aligned}
$$

and so on. Collecting all terms,

$$
\begin{aligned}
\hat{R}(\boldsymbol{\theta})\hat{\Psi}_i\hat{R}(\boldsymbol{\theta})^\dagger &= \hat{\Psi}_i-\frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\tau}\hat{\Psi}_i+\frac{1}{2!\times 4}(\boldsymbol{\theta}\cdot\boldsymbol{\tau})^2\hat{\Psi}_i+\cdots\\
&= \sum_{k=1}^{\infty}\frac{1}{k!}\left(-\frac{i}{2}\boldsymbol{\theta}\cdot\boldsymbol{\tau}\right)^k\hat{\Psi}_i\\
&= \mathcal{R}(\boldsymbol{\theta})\hat{\Psi}_i
\end{aligned}
$$

In last passage, we identified the series as the Taylor expansion of the exponential, recognising the expression for $\mathcal{R}(\boldsymbol{\theta})$ given in Eq. (3.8).

4. Then, the group acts on the product $\hat{\Psi}_i^\dagger\hat{\Psi}_i$ as

$$
\begin{aligned}
\hat{R}\hat{\Psi}_i^\dagger\hat{\Psi}_i\hat{R}^\dagger &= \hat{R}\hat{\Psi}_i^\dagger\hat{R}^\dagger\hat{R}\hat{\Psi}_i\hat{R}^\dagger\\
&= \hat{\Psi}_i^\dagger\mathcal{R}^\dagger\mathcal{R}\hat{\Psi}_i\\
&= \hat{\Psi}_i^\dagger\hat{\Psi}_i
\end{aligned}
$$

being $\mathcal{R}\in\mathrm{SU}(2)$. Recalling Eqns. (3.6), (3.7), both $\hat{H}_V$ and $\hat{N}$ are SU(2) symmetric.

The EHM is symmetric under global spin rotations. Note that $\hat{H}_V$ is actually symmetric under local spin rotations, meaning that we could rotate the spin at each site by a different amount and get the same interaction. The same holds for $\hat{H}_U$. It is the kinetic operator that makes the EHM globally (and not locally) SU(2) symmetric.

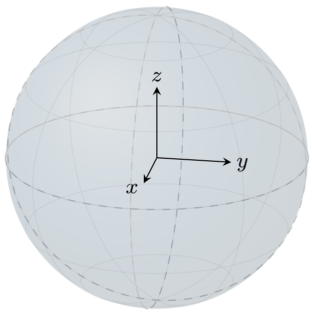
Note that the $U^{\mathrm{c}}(1)$ gauge symmetry and the SU(2) spin rotations symmetry can be combined,

$$
\mathrm{U}^{\mathrm{c}}(1)\otimes\mathrm{SU}(2)=\mathrm{U}(2)
$$

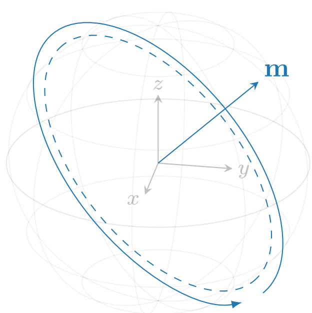
We will refer to U(2) symmetry to indicate full charge-spin conservation. Finally, note the group of 3D rotations contains the subgroup of 2D rotations around a given axis $z$,

$$
\mathrm{U}^z(1)\subset\mathrm{SU}(2)
$$

where the $z$ superscript serves to distinguish from the gauge group. The $z$ axis does not need to be identified with the "real" $z$ axis, perpendicular with the lattice plane. Any given direction works. Inverting our logic, we can build the SU(2) generators, starting from $\hat{S}^z$, after we have chosen the direction $z$. In terms of commutation relations, a given operator can respect $\mathrm{U}^z(1)$ but not SU(2): in that case, it will commutate to zero with $\hat{S}^z$, and not the other spin operators. A graphic rendition of this SSB is given in Fig. 3.2, representing for a given vector $\mathbf{m}$ the passage from full SU(2) rotational symmetry around any direction to reduced U(1) rotational symmetry around just one direction.




a | SU(2) rotations domain before SSB.




b | U(1) rotations domain after SSB.

Figure 3.2 | Graphic representation of the concept of $\mathrm{SU}(2) \to \mathrm{U}^z(1)$ SSB. “Before” SSB, we have full symmetry under any possible rotation (solid color sphere, left panel). “After” SSB we have symmetry only for rotations around some vector $\mathbf{m}$ (dashed tilted circumference, right panel).

### 3.1.3 Particle-hole symmetry at half-filling

The half-filled EHM on the square lattice exhibits particle-hole symmetry (PHS). The presence of this additional symmetry is a consequence of the fact that the square lattice is bipartite. Let us prove this. Consider the generic unitary particle-hole (PH) transformation

$$\mathrm{P} \quad : \quad \hat{c}_{i\sigma} \quad \to \quad e^{i\eta_{i\sigma}} \hat{h}^\dagger_{i\sigma} \tag{3.9}$$

being in general $\eta_i$ a phase dependent on the fermionic quantum state $(i\sigma)$. Inserting the above transformation in the EHM, we get

$$\mathrm{P}\{\hat{H}\} = -t \sum_{\langle ij \rangle} \sum_{\sigma} e^{i(\eta_{i\sigma}+\eta_{j\sigma})} \hat{h}_{i\sigma} \hat{h}^\dagger_{j\sigma} + U \sum_i \hat{h}_{i\uparrow} \hat{h}_{i\downarrow} \hat{h}^\dagger_{i\downarrow} \hat{h}^\dagger_{i\uparrow} - V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}_{i\sigma} \hat{h}_{j\sigma'} \hat{h}^\dagger_{j\sigma'} \hat{h}^\dagger_{i\sigma}$$

For the two interactions $\hat{H}_U$, $\hat{H}_V$ the phase factors cancelled out. We aim to find the conditions under which the above hamiltonian is mapped onto the EHM of (1.2),

$$\mathrm{P}\{\hat{H}\} \stackrel{?}{=} -t \sum_{\langle ij \rangle} \sum_{\sigma} \hat{h}^\dagger_{i\sigma} \hat{h}_{j\sigma} + U \sum_{i \in \mathcal{L}} \hat{h}^\dagger_{i\uparrow} \hat{h}^\dagger_{i\downarrow} \hat{h}_{i\downarrow} \hat{h}_{i\uparrow} - V \sum_{\langle ij \rangle} \sum_{\sigma\sigma'} \hat{h}^\dagger_{i\sigma} \hat{h}_{j\sigma'} \hat{h}_{j\sigma'} \hat{h}_{i\sigma}$$

Before starting the derivation, which is an extension of the argument carried out for the conventional HM [1], let us discuss some properties of the hole operators. Due to the presence of the phase factors in the PHS operative definition, the Fermi anticommutation rules hold regardlessly of the phase factor:

$$\begin{aligned}
\{\hat{h}_{i\sigma}, \hat{h}^\dagger_{j\sigma'}\} &= e^{i(\eta_{i\sigma}-\eta_{j\sigma'})} \{\hat{c}^\dagger_{i\sigma}, \hat{c}_{j\sigma'}\} \\
&= e^{i(\eta_{i\sigma}-\eta_{j\sigma'})} \delta_{ij} \delta_{\sigma\sigma'} \\
&= \delta_{ij} \delta_{\sigma\sigma'}
\end{aligned}$$

The on-site density operator is given by:

$$\hat{h}_{i\sigma} \hat{h}^\dagger_{i\sigma} = \hat{n}_{i\sigma}$$

being $\hat{n}_{i\sigma}$ the fermionic number operator on site $i$ with spin $\sigma$. It follows, for quartic interactions

$$
\begin{aligned}
\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'}\hat{h}^\dagger_{i\sigma} &= -\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'} \\
&= \hat{h}_{i\sigma}\hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'} \\
&= -\hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} + \hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'} \\
&= \hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma} + \hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'} \\
&= -\hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{i\sigma}\hat{h}_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma} + \hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \left(\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma'}\right)\delta_{ij}\delta_{\sigma\sigma'} \\
&= \hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma} + (\hat{n}_{i\sigma} + \hat{n}_{j\sigma'}) - 1 - \left(\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma'}\right)\delta_{ij}\delta_{\sigma\sigma'}
\end{aligned} \tag{3.10}
$$

The first term of the last passage is mirrored with respect to the starting term. Now, let us deal with the kinetic, local and non-local interactions separately.

**Kinetic term.** Consider the kinetic term,

$$\hat{H}_t = -t\sum_{\langle ij\rangle}\sum_{\sigma}\hat{c}^\dagger_{i\sigma}\hat{c}_{j\sigma}$$

Apply the PH transform of Eq. (3.9)

$$\mathrm{P}\{\hat{H}_t\} = -t\sum_{\langle ij\rangle}\sum_{\sigma}e^{i(\eta_{i\sigma}+\eta_{j\sigma})}\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma}$$

Anti-commutate the two fermionic operators, and exchange site labels $i \leftrightarrow j$. We get:

$$\mathrm{P}\{\hat{H}_t\} = -t\sum_{\langle ij\rangle}\sum_{\sigma}e^{i(\eta_{i\sigma}+\eta_{j\sigma}+\pi)}\hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma}$$

A necessary condition on the unitary operation P is the following,

$$\eta_{i\sigma} + \eta_{j\sigma} = (2k-1)\pi \quad \text{with} \quad k \in \mathbb{Z} \tag{3.11}$$

valid for NN sites.

**Local repulsion.** Substitute Eq. (3.10) in $\hat{H}_U$,

$$
\begin{aligned}
\mathrm{P}\{\hat{H}_U\} &= U\sum_{i\in\mathcal{L}}\hat{h}_{i\uparrow}\hat{h}_{i\downarrow}\hat{h}^\dagger_{i\downarrow}\hat{h}^\dagger_{i\uparrow} \\
&= U\sum_{i\in\mathcal{L}}\hat{h}^\dagger_{i\uparrow}\hat{h}^\dagger_{i\downarrow}\hat{h}_{i\downarrow}\hat{h}_{i\uparrow} + U\sum_{i\in\mathcal{L}}(\hat{n}_{i\uparrow}+\hat{n}_{i\downarrow}) - U\sum_{i\in\mathcal{L}}(1) \\
&= U\sum_{i}\hat{h}^\dagger_{i\uparrow}\hat{h}^\dagger_{i\downarrow}\hat{h}_{i\downarrow}\hat{h}_{i\uparrow} + U(\hat{N} - L_xL_y)
\end{aligned}
$$

For the half-filled state $\langle\hat{N}\rangle = L_xL_y$. The unitary map of Eq. (3.9), independently of the condition of Eq. (3.11), leaves the hamiltonian identical at half-filling.

**Non-local attraction.** Proceed identically for $\hat{H}_V$,

$$
\begin{aligned}
\mathrm{P}\{\hat{H}_V\} &= -V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'}\hat{h}^\dagger_{i\sigma} \\
&= -V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma} - V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}(\hat{n}_{i\sigma}+\hat{n}_{j\sigma'}) + V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}(1) \\
&= -V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma} - 2zV(\hat{N} - L_xL_y)
\end{aligned}
$$

| Symmetry | Operations |
|---|---|
| $\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)}$ | One-site discrete translations separately in both directions |
| $\mathrm{C}_4$ | Four-fold point group (real $\pi/2$ angle rotations) |
| $\mathrm{U}^c(1)$ | Charge-conserving global phase shifts |
| $\mathrm{SU}(2)$ | Three-dimensional rotations in global spin space |

Table 3.1 Model symmetries and relative group operations. Note that $\mathrm{U}^c(1)$ represents the gauge symmetry linked to charge conservation.

with $z = 4$ the lattice coordination number. In the last passage, we used that on the square lattice

$$\begin{aligned}\sum_{\langle ij\rangle}(1) &= \sum_{i\in\mathcal{L}/2} z \\ &= \frac{L_x L_y}{2} \times z \end{aligned} \tag{3.12}$$

As before, the non-local attraction is particle-hole symmetric at half-filling. As for the local repulsion, PHS is given just the operation of Eq. (3.9). The condition of Eq. (3.11) is irrelevant.

Recollecting everything, if P is defined by the unitary operation

$$\mathrm{P} \quad : \quad \begin{cases} \hat{c}_{i\sigma} \to \hat{h}^\dagger_{i\sigma} & \text{for} \quad i \in \mathcal{S}_1 \\ \hat{c}_{i\sigma} \to -\hat{h}^\dagger_{i\sigma} & \text{for} \quad i \in \mathcal{S}_2 \end{cases}$$

and the system is half-filled, PHS is respected. Any other phase, as long as it alternates by $\pi$ on the two sublattices, is good enough. As for the conventional HM [1], this result can be generalised to any bipartite lattice. In turn, if a lattice is not bipartite, PHS is not present. To summarise, we report in Tab. 3.1 all identified symmetries.

## 3.2 Symmetry breaking channels in real space

In many-body Physics, a phase transition is described by some order parameter rising from zero to a non-zero value. This is associated with some spontaneous symmetry breaking (SSB). In order to simulate different phases, we need a way to distinguish them based on which symmetries they break (or not). We also need a way to work computationally in a given symmetry sector. HF methods, by reducing the hamiltonian to a non-interacting approximation, offer a simple framework to these ends. Indeed, once we identified all symmetries of a given phase, we just impose them on our approximation of the full hamiltonian. The found ground-state approximation will entail these symmetries by construction.

Recall now the discussion of the HF method of Sec. 2.2. The ending point were Eqns. (2.15), (2.16), which identify the self-consistent single-particle fields which approximate the original quartic hamiltonian. In order to get there, we applied Wick's theorem to the quartic part of the hamiltonian and derived the most general expressions of $\mathbf{h}$. In next sections here apply Wick's theorem to the quartic operators of $\hat{H}_{\mathrm{EHM}}$ and discuss the symmetry properties of the resulting decomposition.

Before starting, let us define precisely the three phases we will investigate in terms of symmetries:

- Normal phase: no broken symmetry.
- Antiferromagnetic phase: simultaneous $\mathrm{SU}(2) \overset{\mathrm{SSB}}{\to} \mathrm{U}^z(1)$ and $\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)} \overset{\mathrm{SSB}}{\to} \mathrm{T}_2^{(x)} \otimes \mathrm{T}_2^{(y)}$. Among all possible realisations of antiferromagnetism, we realise a SDW with alternating distribution of spin densities;
- Superconducting phase: simultaneous $\mathrm{U}^c(1) \overset{\mathrm{SSB}}{\to} 0$ and $\mathrm{C}_4 \overset{\mathrm{SSB}}{\to} \mathrm{C}_2$. We investigate anisotropic superconductivity, allowing for the particle number to be not conserved and for nematic effects.

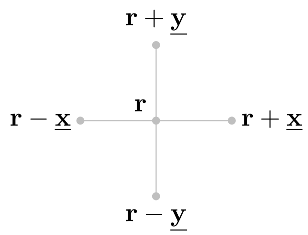



Figure 3.3 | Schematic representation of the four NNs of a given site $i$ for a planar square lattice.

**Nematic effects are observed in unconventional superconductors, and this is the reason of our interest in the possibility of rotational SSBs.** We will theoretically discuss the nematic SSB $C_4 \to C_2$ also for normal and AF phases, as if discrete rotational symmetry was broken. As we will see, while it not trivial to exclude it *a priori*, it will actually turn out that the HF ground state preserves $C_4$ in both cases.

We now move to a more detailed discussion of each hamiltonian contribution. In next sections, we switch from the site index $i$ to the vector notation $\mathbf{r}$, indicating the site 2D position on the lattice

$$i \quad \to \quad \mathbf{r} = (x, y)$$

with $x, y \in \mathbb{N}$. The lattice versors are:

$$\underline{\mathbf{x}} \equiv (1, 0) \qquad \underline{\mathbf{y}} \equiv (0, 1)$$

as in Fig. 3.3.

### 3.2.1 Local repulsion

The local repulsion $\hat{H}_U$ is the quartic operator of the conventional HM,

$$\begin{aligned} \hat{H}_U &= U \sum_{\mathbf{r} \in \mathcal{L}} \hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}\downarrow} \\ &= U \sum_{\mathbf{r} \in \mathcal{L}} \hat{c}^\dagger_{\mathbf{r}\uparrow} \hat{c}^\dagger_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\uparrow} \end{aligned}$$

In order to obtain the self-consistent values of $h^{+-}_{\alpha\beta}$, $h^{++}_{\alpha\beta}$ we need to compute an expectation value of the full hamiltonian. Wick's theorem guarantees that, for the wavefunctions we are using, the expectation value of a quartic fermionic operator is the sum of all fully contracted terms. Recalling the nomenclature of Eq. (2.3),

$$\langle \hat{c}^\dagger_{\mathbf{r}\uparrow} \hat{c}^\dagger_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\uparrow} \rangle = \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow} \hat{c}_{\mathbf{r}\uparrow} \rangle \langle \hat{c}^\dagger_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\downarrow} \rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow} \hat{c}_{\mathbf{r}\downarrow} \rangle \langle \hat{c}^\dagger_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\uparrow} \rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow} \hat{c}^\dagger_{\mathbf{r}\downarrow} \rangle \langle \hat{c}_{\mathbf{r}\downarrow} \hat{c}_{\mathbf{r}\uparrow} \rangle}_{\text{Cooper}} \tag{3.13}$$

where the equal sign holds as long as we perform the averaging operation on the subspace of **gaussian states**. Now we discuss the above EVs separately, identifying if any can be neglected based on selection rules in a given phase.

**Hartree term.** The Hartree part of the above decomposition is a product of density EVs,

$$\langle \hat{n}_{\mathbf{r}\uparrow} \rangle \langle \hat{n}_{\mathbf{r}\downarrow} \rangle$$

Recalling the groups of Tab. 3.1, we see that the density operator preserves all symmetries. Single-site density operators commutate with spin operators and are gauge-invariant. Moreover, shifting or rotating the lattice leaves density operators unchanged. Thus no selection rule applies, and the above EVs are in principle non-zero.

**Fock terms.** For the Fock part, notice that

$$\hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\downarrow} = \hat{S}^{+}_{\mathbf{r}}$$

being $S^{+}_{\mathbf{r}}$ the local raising operator. From SU(2) algebra we have:

$$\left[\hat{S}^{+}_{\mathbf{r}}, \hat{S}^{\alpha}\right] \neq 0 \qquad \alpha \in \{x, y, z\}$$

If the ground-state is invariant under global spin rotations, these EVs must be zero. Gauge symmetry gives no selection rules, due to Eq. (3.5).

**Cooper terms.** The Cooper part can be non-zero only if charge is not conserved. In terms of the gauge symmetry, it suffices to apply the map of Eq. (3.4) to $\hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}^{\dagger}_{\mathbf{r}\downarrow}$

$$\hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}^{\dagger}_{\mathbf{r}\downarrow} \quad \to \quad e^{2i\eta}\hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}^{\dagger}_{\mathbf{r}\downarrow}$$

to conclude that this operator is in general not gauge invariant. It is instead SU(2)-symmetric. To prove this, it suffices to compute commutators of $\hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}^{\dagger}_{\mathbf{r}\downarrow}$ with $\hat{S}^{z}$, $\hat{S}^{+}$:

$$\begin{aligned}
\left[\hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}^{\dagger}_{\mathbf{r}\downarrow}, \hat{S}^{z}\right] &= \left[\hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}^{\dagger}_{\mathbf{r}\downarrow}, \frac{1}{2}\sum_{\mathbf{r}'\in\mathcal{L}}(\hat{n}_{\mathbf{r}'\uparrow} - \hat{n}_{\mathbf{r}'\downarrow})\right] \\
&= \frac{1}{2}\left[\hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}^{\dagger}_{\mathbf{r}\downarrow}, \hat{n}_{\mathbf{r}\uparrow} - \hat{n}_{\mathbf{r}\downarrow}\right] \\
&= \frac{1}{2}\left[\hat{c}^{\dagger}_{\mathbf{r}\uparrow}, \hat{n}_{\mathbf{r}\uparrow}\right]\hat{c}^{\dagger}_{\mathbf{r}\downarrow} - \frac{1}{2}\hat{c}^{\dagger}_{\mathbf{r}\uparrow}\left[\hat{c}^{\dagger}_{\mathbf{r}\downarrow}, \hat{n}_{\mathbf{r}\downarrow}\right]
\end{aligned}$$

Since:

$$\begin{aligned}
\left[\hat{c}^{\dagger}_{\mathbf{r}\sigma}, \hat{n}_{\mathbf{r}\sigma}\right] &= \hat{c}^{\dagger}_{\mathbf{r}\sigma}\hat{c}^{\dagger}_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma} - \hat{c}^{\dagger}_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma}\hat{c}^{\dagger}_{\mathbf{r}\sigma} \\
&= -\hat{c}^{\dagger}_{\mathbf{r}\sigma}\left(1 - \hat{c}^{\dagger}_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma}\right) \\
&= -\hat{c}^{\dagger}_{\mathbf{r}\sigma}
\end{aligned} \tag{3.14}$$

we have:

$$\left[\hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}^{\dagger}_{\mathbf{r}\downarrow}, \hat{S}^{z}\right] = -\frac{1}{2}\hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}^{\dagger}_{\mathbf{r}\downarrow} + \frac{1}{2}\hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}^{\dagger}_{\mathbf{r}\downarrow} = 0$$

Moreover, considering the raising operator:

$$\begin{aligned}
\left[\hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}^{\dagger}_{\mathbf{r}\downarrow}, \hat{S}^{+}\right] &= \left[\hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}^{\dagger}_{\mathbf{r}\downarrow}, \frac{1}{2}\sum_{\mathbf{r}'\in\mathcal{L}}\hat{c}^{\dagger}_{\mathbf{r}'\uparrow}\hat{c}_{\mathbf{r}'\downarrow}\right] \\
&= \left[\hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}^{\dagger}_{\mathbf{r}\downarrow}, \hat{c}^{\dagger}_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\downarrow}\right] \\
&= \left[\hat{c}^{\dagger}_{\mathbf{r}\uparrow}, \hat{c}^{\dagger}_{\mathbf{r}\uparrow}\right]\hat{c}^{\dagger}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow} - \left(\hat{c}^{\dagger}_{\mathbf{r}\uparrow}\right)^{2}\left[\hat{c}^{\dagger}_{\mathbf{r}\downarrow}, \hat{c}_{\mathbf{r}\downarrow}\right] \\
&= 0
\end{aligned}$$

where we used the facts that every operator commutates with itself, and a squared creation operator is null. These commutation relations, together, prove that Cooper EVs are legitimately non-zero for a SU(2)-preserving phase.

In Tab. 3.2 we collect the selection rules for the Hartree, Fock and Cooper terms of Eq. (3.13), based on the symmetry specifications given for each phase in above paragraphs. Note that Fock-like EVs are prohibited for all phases.

| Term | Normal phase | Antiferromagnet | Superconductor |
|---|---|---|---|
| Hartree | | | |
| Fock | Prohibited | Prohibited | Prohibited |
| Cooper | Prohibited | Prohibited | |

Table 3.2 Summary of selection rules for the Wick's decomposition of Eq. (3.13) of the local repulsion $\hat{H}_U$. An empty entry indicates no selection rule.

### 3.2.2 Non-local attraction

In order to analyse similarly the non-local attraction $\hat{H}_V$, it is useful to decompose the operator in spin channels:

$$\hat{H}_V \equiv \underbrace{\hat{H}_V^{\uparrow\uparrow} + \hat{H}_V^{\downarrow\downarrow}}_{\text{Same spin}} + \underbrace{\hat{H}_V^{\uparrow\downarrow} + \hat{H}_V^{\downarrow\uparrow}}_{\text{Opposite spin}}$$

being

$$\hat{H}_V^{\sigma\sigma'} \equiv -V \sum_{\langle \mathbf{r},\mathbf{r}'\rangle} \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}'\sigma'}$$

Let us start from the opposite spin sector. On the square lattice (as for any bipartite lattice) NNs belong to different sublattices. This implies that any NNs sum can be rewritten as:

$$\sum_{\langle \mathbf{r}\mathbf{r}'\rangle} f(\mathbf{r},\mathbf{r}') = \sum_{\mathbf{r}\in\mathcal{L}/2} \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} f(\mathbf{r},\mathbf{r}+\boldsymbol{\delta}) \tag{3.15}$$

with $f$ some function. We are using the notation of Fig. 3.3. The symbol "$\mathcal{L}/2$" indicates one of the two sublattices $\mathcal{S}_1$, $\mathcal{S}_2$ represented in Fig. 3.1. We can rewrite:

$$\hat{H}_V^{\sigma\sigma'} = -V \sum_{\mathbf{r}\in\mathcal{L}/2} \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \left(\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma'} + \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\sigma'}\right)$$

Now, if we choose $\mathcal{L}/2 = \mathcal{S}_1$, this gives:

$$\hat{H}_V^{\uparrow\downarrow} = -V \sum_{\mathbf{r}\in\mathcal{S}_1} \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \left(\hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\downarrow} + \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\downarrow}\right)$$
$$\hat{H}_V^{\downarrow\uparrow} = -V \sum_{\mathbf{r}\in\mathcal{S}_1} \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \left(\hat{n}_{\mathbf{r}\downarrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\uparrow} + \hat{n}_{\mathbf{r}\downarrow}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\uparrow}\right)$$

These two operators are "complementary": $\sigma =\uparrow$ densities act on $\mathcal{S}_1$ for $\hat{H}_V^{\uparrow\downarrow}$, on $\mathcal{S}_2$ for $\hat{H}_V^{\downarrow\uparrow}$; $\sigma =\downarrow$ densities, the opposite. Then their sum can be expressed as:

$$\begin{aligned}\hat{H}_V^{(\text{o.s.})} &\equiv \hat{H}_V^{\uparrow\downarrow} + \hat{H}_V^{\downarrow\uparrow} \\ &= -V \sum_{\mathbf{r}\in\mathcal{L}} \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \left(\hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\downarrow} + \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\downarrow}\right)\end{aligned} \tag{3.16}$$

Wick's theorem applied to this quartic operator gives:

$$\begin{aligned}\langle \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\rangle &= \langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle \\ &\simeq \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\rangle\langle \hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Cooper}}\end{aligned} \tag{3.17}$$

For the same-spin sector we define

$$\begin{aligned}\hat{H}_V^{(\text{s.s.})} &\equiv \hat{H}_V^{\uparrow\uparrow} + \hat{H}_V^{\downarrow\downarrow} \\ &= -V \sum_{\mathbf{r}\in\mathcal{L}/2} \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \sum_{\sigma\in\{\uparrow,\downarrow\}} \left(\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma} + \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\sigma}\right)\end{aligned} \tag{3.18}$$

| Sector | Term | Normal phase | Antiferromagnet | Superconductor |
|---|---|---|---|---|
| o.s. | Hartree | | | |
| | Fock | Prohibited | Prohibited | Prohibited |
| | Cooper | Prohibited | Prohibited | |
| s.s. | Hartree | | | |
| | Fock | | | |
| | Cooper | Prohibited | Prohibited | |

Table 3.3 Summary of selection rules for the Wick's decomposition of Eqns. (3.17), (3.19) of the non-local attraction $\hat{H}_V$. An empty entry indicates no selection rule.

Note that, with respect to the o.s. sector, we are summing on half the sites two times: one for spin $\uparrow$, one for spin $\downarrow$. Applying Wick's theorem:

$$\begin{aligned}\langle \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\rangle &= \langle \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\hat{c}_{\mathbf{r}\sigma}\rangle \\ &\simeq \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\sigma}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\rangle}_{\text{Hartree}} - \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\rangle\langle \hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\hat{c}_{\mathbf{r}\sigma}\rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\rangle\langle \hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma}\hat{c}_{\mathbf{r}\sigma}\rangle}_{\text{Cooper}}\end{aligned} \tag{3.19}$$

As we did for $\hat{H}_U$, we analyse each separately.

**Hartree terms.** As it was for $\hat{H}_U$, Hartree EVs of Eqns. (3.16), (3.18) are not subject to selection rules for any symmetry of Tab. 3.1. Thus they are allowed in both sectors.

**Fock terms.** The Fock EVs from the two sectors behave differently. Let us compute the following generic commutation relations:

$$\begin{aligned}\left[\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \hat{S}^z\right] &= \left[\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \frac{1}{2}\sum_{\mathbf{r}'\in\mathcal{L}}(\hat{n}_{\mathbf{r}'\uparrow} - \hat{n}_{\mathbf{r}'\downarrow})\right] \\ &= \frac{(-1)^{\delta_{\sigma=\downarrow}}}{2}\left[\hat{c}^\dagger_{\mathbf{r}\sigma}, \hat{n}_{\mathbf{r}\sigma}\right]\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'} + \frac{(-1)^{\delta_{\sigma'=\downarrow}}}{2}\hat{c}^\dagger_{\mathbf{r}\sigma}\left[\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma}\right] \\ &= \left(-\frac{(-1)^{\delta_{\sigma=\downarrow}}}{2} + \frac{(-1)^{\delta_{\sigma'=\downarrow}}}{2}\right)\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}\end{aligned} \tag{3.20}$$

where in last passage we used Eq. (3.14) and its conjugate version,

$$[\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}, \hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma}] = \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\sigma'}$$

Now, if $\sigma = \sigma'$ the last passage of Eq. (3.20) is zero; otherwise is not. This proves that the o.s. Fock EVs are non-zero only for a phase that completely breaks SU(2); instead, in the s.s. sector we can have non-zero EVs for a phase that reduces SU(2) to $\mathrm{U}^z(1)$.

**Cooper terms.** For the Cooper terms, identical considerations as for $\hat{H}_U$ hold. Gauge symmetry must be broken to allow for non zero EVs, while SU(2) symmetry survives. The proof is analogous to the one given for $\hat{H}_U$.

In Tab. 3.3 we summarise these considerations, giving the selection rules on the EVs of Eqns. (3.17), (3.19), dividing in the two sectors "o.s." and "s.s.". By identifying these selection rules, we set grounds for the detailed derivation of each chapter and heavily simplify application of HF model to the EHM. We now move to the last part of this chapter: reciprocal space description of the EHM.

## 3.3 The EHM in reciprocal space

The EHM is symmetric under discrete lattice translations. We will investigate phases that do not (completely) break translational invariance. It is thus convenient to move to reciprocal space, by

Fourier-transforming the hamiltonian. We will use the following convention:

$$\hat{c}_{\mathbf{r}\sigma} \stackrel{\mathcal{F}}{=} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}\in \mathrm{BZ}} e^{-i\mathbf{k}\cdot\mathbf{r}} \hat{c}_{\mathbf{k}\sigma} \tag{3.21}$$

Throughout the text, $\stackrel{\mathcal{F}}{=}$ will indicate a passage where a Fourier transform is used. "BZ" is shorthand notation for the Brillouin Zone.

It is useful to recall that the following equation for Kronecker's delta holds:

$$\frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}} e^{i\mathbf{k}\cdot\mathbf{r}} = \delta_{\mathbf{k}=\mathbf{0}} \tag{3.22}$$

On one sublattice:

$$\frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}/2} e^{i\mathbf{k}\cdot\mathbf{r}} = \frac{1}{2}\delta_{\mathbf{k}=\mathbf{0}} \tag{3.23}$$

The notation $\mathcal{L}/2$ is shorthand to indicate $\mathcal{S}_1$ or $\mathcal{S}_2$.

**Kinetic part.** Let us transform $\hat{H}_t$:

$$\begin{aligned}
\hat{H}_t &= -t \sum_{\langle \mathbf{r}\mathbf{r}'\rangle} \sum_{\sigma} \left[\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}'\sigma} + \text{h.c.}\right] \\
&\stackrel{\mathcal{F}}{=} -t \sum_{\langle \mathbf{r}\mathbf{r}'\rangle} \sum_{\sigma} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}\in\mathrm{BZ}} e^{i\mathbf{k}\cdot\mathbf{r}} \hat{c}^\dagger_{\mathbf{k}\sigma} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}'\in\mathrm{BZ}} e^{-i\mathbf{k}'\cdot\mathbf{r}'} \hat{c}_{\mathbf{k}'\sigma} + \text{h.c.} \\
&= -2t \sum_{\mathbf{k},\mathbf{k}'} \sum_{\sigma} \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}'\sigma} \times \frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}/2} e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}} \times \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} e^{i\mathbf{k}'\cdot\boldsymbol{\delta}} \\
&= -2t \sum_{\mathbf{k},\mathbf{k}'} \sum_{\sigma} \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} \left[\frac{e^{ik_x} + e^{-ik_x}}{2} + \frac{e^{ik_y} + e^{-ik_y}}{2}\right] &(3.24) \\
&= \sum_{\mathbf{k}\in\mathrm{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}} \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} &(3.25)
\end{aligned}$$

where we used Eqns. (3.15), (3.23). We defined the bare bands:

$$\epsilon_{\mathbf{k}} \equiv -2t\left(\cos k_x + \cos k_y\right) \tag{3.26}$$

**Local repulsion.** The local repulsion transforms as:

$$\begin{aligned}
\hat{H}_U &= U \sum_{\mathbf{r}\in\mathcal{L}} \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\downarrow}\hat{c}_{\mathbf{r}\uparrow} \\
&\stackrel{\mathcal{F}}{=} U \sum_{\mathbf{r}\in\mathcal{L}} \frac{1}{(L_x L_y)^2} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} e^{i\mathbf{k}_1\cdot\mathbf{r}}\hat{c}^\dagger_{\mathbf{k}_1\uparrow} e^{i\mathbf{k}_2\cdot\mathbf{r}}\hat{c}^\dagger_{\mathbf{k}_2\downarrow} e^{-i\mathbf{k}_3\cdot\mathbf{r}}\hat{c}_{\mathbf{k}_3\downarrow} e^{-i\mathbf{k}_4\cdot\mathbf{r}}\hat{c}_{\mathbf{k}_4\uparrow} \\
&= \frac{U}{L_x L_y} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} \hat{c}^\dagger_{\mathbf{k}_1\uparrow}\hat{c}^\dagger_{\mathbf{k}_2\downarrow}\hat{c}_{\mathbf{k}_3\downarrow}\hat{c}_{\mathbf{k}_4\uparrow} \times \frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{L}} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} \\
&= \frac{U}{L_x L_y} \sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} \hat{c}^\dagger_{\mathbf{k}_1\uparrow}\hat{c}^\dagger_{\mathbf{k}_2\downarrow}\hat{c}_{\mathbf{k}_3\downarrow}\hat{c}_{\mathbf{k}_4\uparrow}\delta_{\mathbf{k}_1+\mathbf{k}_2=\mathbf{k}_3+\mathbf{k}_4}
\end{aligned}$$

where we used Eq. (3.22). Let $\mathbf{k}_1 + \mathbf{k}_2 = \mathbf{k}_3 + \mathbf{k}_4 = \mathbf{K}$, and define $\mathbf{k}$, $\mathbf{k}'$ such that

$$\mathbf{k}_1 \equiv \mathbf{K} + \mathbf{k} \qquad \mathbf{k}_2 \equiv \mathbf{K} - \mathbf{k} \qquad \mathbf{k}_3 \equiv \mathbf{K} - \mathbf{k}' \qquad \mathbf{k}_4 \equiv \mathbf{K} + \mathbf{k}'$$

Sums over these variables must be intended as over the Brillouin Zone (BZ). Hence:

$$\hat{H}_U \stackrel{\mathcal{F}}{=} \frac{U}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow}\hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \tag{3.27}$$

**Non-local attraction.** To transform the non-local attraction $\hat{H}_V$, it is useful to start by defining:

$$h_{\mathbf{r}\sigma\sigma'} \equiv -V \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \left(\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma'} + \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\sigma'}\right)$$

with:

$$\hat{H}_V = \sum_{\mathbf{r}\in\mathcal{L}/2}\sum_{\sigma,\sigma'} h_{\mathbf{r}\sigma\sigma'}$$

The product of densities on neighbouring sites transforms as:

$$\begin{aligned}\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'} &= \hat{c}^{\dagger}_{\mathbf{r}\sigma}\hat{c}^{\dagger}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'}\hat{c}_{\mathbf{r}\sigma}\\ &\overset{\mathcal{F}}{=} \frac{1}{(L_xL_y)^2}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}}e^{\pm i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}}\hat{c}^{\dagger}_{\mathbf{k}_1\sigma}\hat{c}^{\dagger}_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma}\end{aligned}$$

Then:

$$\begin{aligned}h_{\mathbf{r}\sigma\sigma'} &\overset{\mathcal{F}}{=} -\frac{V}{(L_xL_y)^2}\sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}}\\ &\qquad\times\left(e^{i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}}+e^{-i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}}\right)\hat{c}^{\dagger}_{\mathbf{k}_1\sigma}\hat{c}^{\dagger}_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma}\\ &= -\frac{2V}{(L_xL_y)^2}\sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}}\sum_{\mathbf{k}_1,\mathbf{k}_2,\mathbf{k}_3,\mathbf{k}_4} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}}\cos\left[(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}\right]\hat{c}^{\dagger}_{\mathbf{k}_1\sigma}\hat{c}^{\dagger}_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma}\end{aligned}$$

The full non-local interaction is given by summing over all sites of $\mathcal{L}/2$ (which is, one sublattice). Applying Eq. (3.23) gives back momentum conservation

$$\frac{1}{L_xL_y}\sum_{\mathbf{r}\in\mathcal{L}/2} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} = \frac{1}{2}\delta_{\mathbf{k}_1+\mathbf{k}_2=\mathbf{k}_3+\mathbf{k}_4}$$

where we used Eq. (3.23). Define $\mathbf{k}$, $\mathbf{k}'$ and $\mathbf{K}$ as we did fo the local repulsion, and

$$\delta\mathbf{k} \equiv \mathbf{k}-\mathbf{k}'$$

Hence:

$$\begin{aligned}\hat{H}_V &= \sum_{\mathbf{r}\in\mathcal{L}/2}\sum_{\sigma,\sigma'} h_{\mathbf{r}\sigma\sigma'}\\ &\overset{\mathcal{F}}{=} -\frac{V}{L_xL_y}\sum_{\sigma\sigma'}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\cos\left(\delta\mathbf{k}\cdot\boldsymbol{\delta}\right)\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma'}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma'}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}\\ &= -\frac{V}{L_xL_y}\sum_{\sigma\sigma'}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma'}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma'}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}\end{aligned}\tag{3.28}$$

Finally, we can separate in same spin and opposite spin channels. This gives:

$$\begin{aligned}\hat{H}_V \overset{\mathcal{F}}{=} &\overbrace{-\frac{V}{L_xL_y}\sum_{\sigma}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}}^{\text{s.s.}}\\ &\underbrace{-\frac{V}{L_xL_y}\sum_{\sigma}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}}_{\text{o.s.}}\end{aligned}$$

The o.s. part can be simplified: since

$$\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} = \hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}$$

due to fermionic anti-commutation rules, and since the composite transform

$$\mathbf{k} \to -\mathbf{k} \qquad \mathbf{k}' \to -\mathbf{k}'$$

leaves the form factor $\cos(\delta k_x) + \cos(\delta k_y)$ invariant, we reduce the spin sum to two times the $\sigma = \uparrow$ part,

$$\hat{H}_V^{(\text{o.s.})} = -\frac{2V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \left[\cos(\delta k_x) + \cos(\delta k_y)\right] \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \tag{3.29}$$

The $\mathcal{F}$-transform of the s.s. part is given explicitly by:

$$\begin{aligned} \hat{H}_V^{(\text{s.s.})} = -\frac{V}{L_x L_y} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} &\left[\cos(\delta k_x) + \cos(\delta k_y)\right] \\ &\times \left[\hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\uparrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\uparrow} \hat{c}_{\mathbf{K}-\mathbf{k}'\uparrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} + \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\downarrow} \hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\downarrow} \hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow} \hat{c}_{\mathbf{K}+\mathbf{k}'\downarrow}\right] \end{aligned} \tag{3.30}$$

All these expressions are going to be used in the analysis of each phase. In reciprocal space, for a quartic operator $\hat{c}^\dagger_{\mathbf{k}_1} \hat{c}^\dagger_{\mathbf{k}_2} \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4}$ we distinguish contractions as follows:

- Hartree contractions:

$$\wick{\c1{\hat{c}}^\dagger_{\mathbf{k}_1} \hat{c}^\dagger_{\mathbf{k}_2} \hat{c}_{\mathbf{k}_3} \c1{\hat{c}}_{\mathbf{k}_4}} \qquad \wick{\hat{c}^\dagger_{\mathbf{k}_1} \c1{\hat{c}}^\dagger_{\mathbf{k}_2} \c1{\hat{c}}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4}}$$

- Fock contractions:

$$\wick{\c1{\hat{c}}^\dagger_{\mathbf{k}_1} \hat{c}^\dagger_{\mathbf{k}_2} \c1{\hat{c}}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4}} \qquad \wick{\hat{c}^\dagger_{\mathbf{k}_1} \c1{\hat{c}}^\dagger_{\mathbf{k}_2} \hat{c}_{\mathbf{k}_3} \c1{\hat{c}}_{\mathbf{k}_4}}$$

- Cooper contractions:

$$\wick{\c1{\hat{c}}^\dagger_{\mathbf{k}_1} \c1{\hat{c}}^\dagger_{\mathbf{k}_2} \hat{c}_{\mathbf{k}_3} \hat{c}_{\mathbf{k}_4}} \qquad \wick{\hat{c}^\dagger_{\mathbf{k}_1} \hat{c}^\dagger_{\mathbf{k}_2} \c1{\hat{c}}_{\mathbf{k}_3} \c1{\hat{c}}_{\mathbf{k}_4}}$$

with appropriate sign factors. With the EHM expressed in reciprocal space, there is no general feature left to address. We can move to the detailed discussion of each phase.

## 3.4 Square lattice spatial symmetry channels

This short conclusive section is apparently rather uncorrelated with all we said in the previous parts. We here introduce a set of orthonormal functions that results particularly useful to represent quantities defined on NNs sites on the square lattice. Let us start from a generic function of two variables $f(\mathbf{r}, \mathbf{r}')$. Suppose that, for a given centre $\mathbf{r}$, the function is non-zero only on the site itself ($\mathbf{r}' = \mathbf{r}$) and its NNs $\mathbf{r}' = \mathbf{r} \pm \boldsymbol{\delta}$, with $\boldsymbol{\delta} \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\}$. In other words,

$$f(\mathbf{r}, \mathbf{r}') \neq 0 \qquad \Longleftrightarrow \qquad |\mathbf{r} - \mathbf{r}'| = 0, 1$$

in units of the lattice step. The following identity holds:

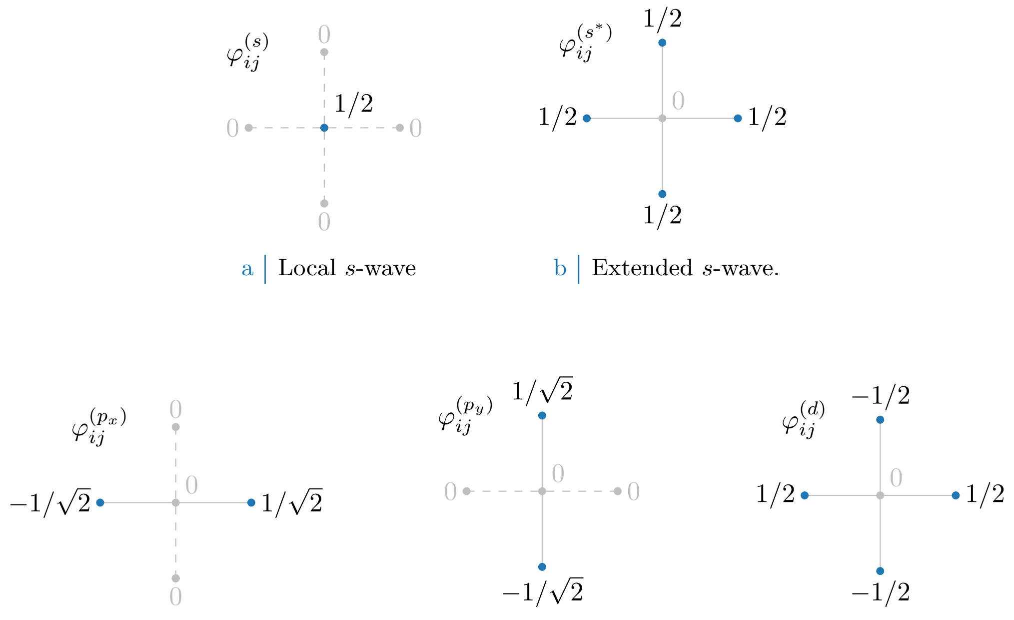
$$\begin{aligned} f(\mathbf{r}, \mathbf{r} + \underline{\mathbf{x}}) &= f(\mathbf{r}, \mathbf{r}')\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} \\ &= f(\mathbf{r}, \mathbf{r}') \left(\varphi^{(s^*)}_{\mathbf{r}\mathbf{r}'} + \varphi^{(d)}_{\mathbf{r}\mathbf{r}'} + \sqrt{2}\varphi^{(p_x)}_{\mathbf{r}\mathbf{r}'}\right) \end{aligned}$$

where we defined the set of orthonormal functions $\varphi^{(\gamma)}_{\mathbf{r}\mathbf{r}'}$ listed explicitly in Tab. 3.4 and sketched in Fig. 3.4. The orthonormality condition reads:

$$\frac{1}{L_x L_y} \sum_{\mathbf{r}\mathbf{r}'} \varphi^{(\gamma)}_{\mathbf{r}\mathbf{r}'} \varphi^{(\gamma')*}_{\mathbf{r}\mathbf{r}'} = \delta_{\gamma\gamma'}$$

Similarly, also the following relations hold:

$$\begin{aligned} f(\mathbf{r}, \mathbf{r}) &= f(\mathbf{r}, \mathbf{r}')\varphi^{(s)}_{\mathbf{r}\mathbf{r}'} \\ f(\mathbf{r}, \mathbf{r} - \underline{\mathbf{x}}) &= f(\mathbf{r}, \mathbf{r}') \left(\varphi^{(s^*)}_{\mathbf{r}\mathbf{r}'} + \varphi^{(d)}_{\mathbf{r}\mathbf{r}'} - \sqrt{2}\varphi^{(p_x)}_{\mathbf{r}\mathbf{r}'}\right) \\ f(\mathbf{r}, \mathbf{r} + \underline{\mathbf{y}}) &= f(\mathbf{r}, \mathbf{r}') \left(\varphi^{(s^*)}_{\mathbf{r}\mathbf{r}'} - \varphi^{(d)}_{\mathbf{r}\mathbf{r}'} + \sqrt{2}\varphi^{(p_y)}_{\mathbf{r}\mathbf{r}'}\right) \\ f(\mathbf{r}, \mathbf{r} - \underline{\mathbf{y}}) &= f(\mathbf{r}, \mathbf{r}') \left(\varphi^{(s^*)}_{\mathbf{r}\mathbf{r}'} - \varphi^{(d)}_{\mathbf{r}\mathbf{r}'} - \sqrt{2}\varphi^{(p_y)}_{\mathbf{r}\mathbf{r}'}\right) \end{aligned}$$




a | Local $s$-wave

b | Extended $s$-wave.

c | $p_x$-wave.

d | $p_y$-wave.

e | $d_{x^2-y^2}$-wave.

Figure 3.4 | Form factors at different topologies, as listed in Tab. 3.4. In figures five sites are represented: the hub site $i$ and its four NN. Solid lines represent non-zero values for $\varphi_{\boldsymbol{\delta}}$, while dashed lines represent vanishing factors.

| **Structure** | **Form factor** | **Graph** |
|---|---|---|
| $s$-wave | $\varphi_{\mathbf{r}\mathbf{r}'}^{(s)} = \delta_{\mathbf{r}\mathbf{r}'}$ | Fig. 3.4a |
| Extended $s$-wave | $\varphi_{\mathbf{r}\mathbf{r}'}^{(s^*)} = \dfrac{1}{2}\left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}}\right]$ | Fig. 3.4b |
| $p_x$-wave | $\varphi_{\mathbf{r}\mathbf{r}'}^{(p_x)} = \dfrac{1}{\sqrt{2}}\left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} - \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}}\right]$ | Fig. 3.4c |
| $p_y$-wave | $\varphi_{\mathbf{r}\mathbf{r}'}^{(p_y)} = \dfrac{1}{\sqrt{2}}\left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} - \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}}\right]$ | Fig. 3.4d |
| $d_{x^2-y^2}$-wave | $\varphi_{\mathbf{r}\mathbf{r}'}^{(d)} = \dfrac{1}{2}\left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} - \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} - \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}}\right]$ | Fig. 3.4e |

Table 3.4 | Form factors for a function defined on a square lattice, with a prefactor included for normalization. We use lattice vector notation here. The last column indicates the graph representation of each contribution given in Fig. 3.4. Subscript $x^2 - y^2$ is omitted for notational clarity.

Thus, defining:

$$
\begin{aligned}
f^{(s)} &\equiv f(\mathbf{r},\mathbf{r}) \\
f^{(s^*)} &\equiv f(\mathbf{r},\mathbf{r}+\underline{\mathbf{x}}) + f(\mathbf{r},\mathbf{r}-\underline{\mathbf{x}}) + f(\mathbf{r},\mathbf{r}+\underline{\mathbf{y}}) + f(\mathbf{r},\mathbf{r}-\underline{\mathbf{y}}) \\
f^{(p_x)} &\equiv f(\mathbf{r},\mathbf{r}+\underline{\mathbf{x}}) - f(\mathbf{r},\mathbf{r}-\underline{\mathbf{x}}) \\
f^{(p_y)} &\equiv f(\mathbf{r},\mathbf{r}+\underline{\mathbf{y}}) - f(\mathbf{r},\mathbf{r}-\underline{\mathbf{y}}) \\
f^{(d)} &\equiv f(\mathbf{r},\mathbf{r}+\underline{\mathbf{x}}) + f(\mathbf{r},\mathbf{r}-\underline{\mathbf{x}}) - f(\mathbf{r},\mathbf{r}+\underline{\mathbf{y}}) - f(\mathbf{r},\mathbf{r}-\underline{\mathbf{y}})
\end{aligned}
$$

we get the decomposition:

$$
f(\mathbf{r},\mathbf{r}') = \sum_{\gamma} f^{(\gamma)} \varphi_{\mathbf{r}\mathbf{r}'}^{(\gamma)}
$$

| Structure | Structure factor | Graph |
|---:|---|---|
| $s$-wave | $\varphi_{\mathbf{k}}^{(s)} = 1$ | Fig. 3.4a |
| Extended $s$-wave | $\varphi_{\mathbf{k}}^{(s^*)} = \cos k_x + \cos k_y$ | Fig. 3.4b |
| $p_x$-wave | $\varphi_{\mathbf{k}}^{(p_x)} = i\sqrt{2}\sin k_x$ | Fig. 3.4c |
| $p_y$-wave | $\varphi_{\mathbf{k}}^{(p_y)} = i\sqrt{2}\sin k_y$ | Fig. 3.4d |
| $d_{x^2-y^2}$-wave | $\varphi_{\mathbf{k}}^{(d)} = \cos k_x - \cos k_y$ | Fig. 3.4e |

Table 3.5 Structure factors derived from the correlation structures of Tab. 3.4. The functions hereby defined are orthonormal, and are plotted in Fig. 3.5.

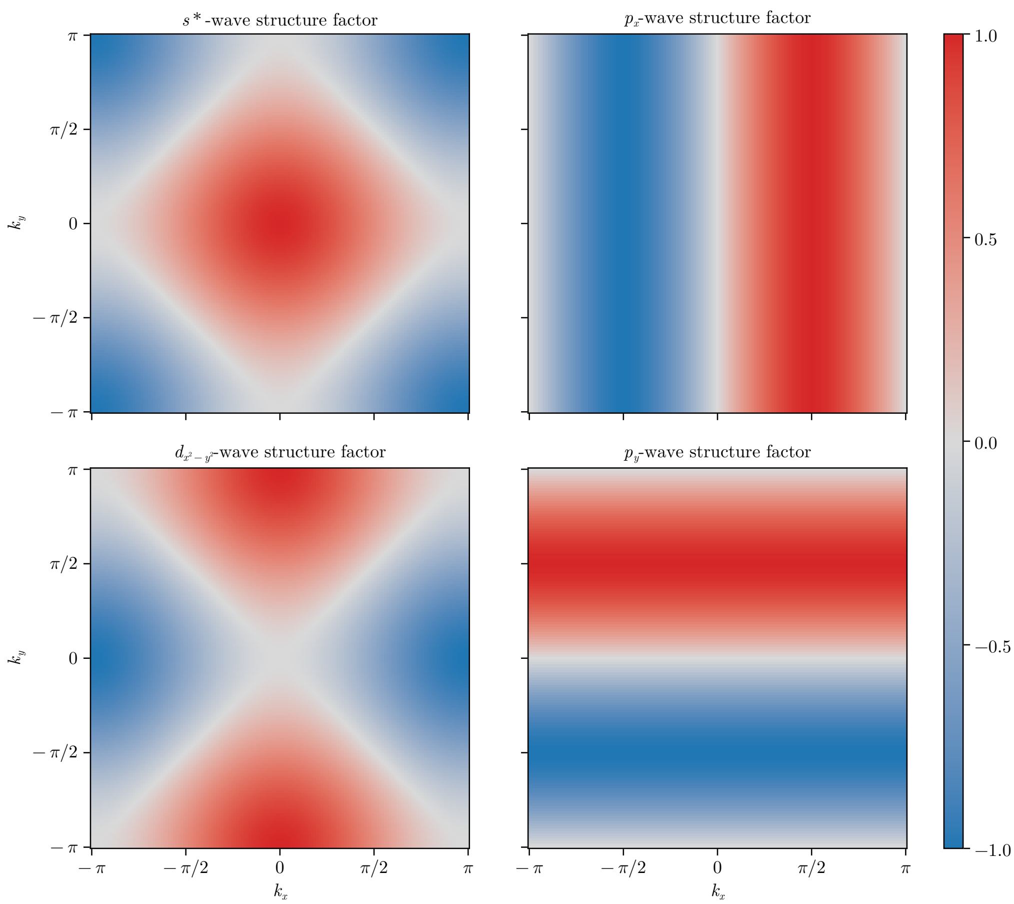



Figure 3.5 Representation of the various structure factors listed in Tab. 3.5 on the BZ. For the $s^*$ and $d_{x^2-y^2}$-wave factors, the real part is plotted; for the $p_\ell$ factors, the imaginary part is plotted.

We decomposed the original function in its harmonics. Moving to reciprocal space, we get the $\mathcal{F}$-transformed functions of Tab. 3.5, also satisfying the orthonormality condition:

$$\frac{1}{L_x L_y}\sum_{\mathbf{k}} \varphi_{\mathbf{k}}^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma')*} = \delta_{\gamma\gamma'}$$

The functions of Tab. 3.5 are plotted in the BZ in Fig. 3.5.

We will refer to these harmonics as “symmetry channels”, indicating the various spatial components of the order parameter associated to a given phase transition. As we will see, indeed, we consider order parameters that have specific transformation properties under planar rotations.

To distinguish the various spatial harmonics of the order parameter is useful in order to apply symmetry-based selection rules.

Finally, a very useful mathematical identity is given by the following argument. Consider Eq. (3.28), where is present the factor $\cos(\delta k_x)+\cos(\delta k_y)$ (with $\delta k_x = k_x - k'_x$ and $\delta k_y = k_y - k'_y$). Expanding,

$$\cos(\delta k_x) + \cos(\delta k_y) = c_x c'_x + c_y c'_y + s_x s'_x + s_y s'_y$$

In last passage, for the sake of readability, we used the notations

$$c_\ell \equiv \cos k_\ell \qquad s_\ell \equiv \sin k_\ell \qquad c'_\ell \equiv \cos k'_\ell \qquad s'_\ell \equiv \sin k'_\ell$$

Group the four terms,

$$\underbrace{\left(c_x c'_x + c_y c'_y\right)}_{\text{Symmetric}} + \underbrace{\left(s_x s'_x + s_y s'_y\right)}_{\text{Anti-symmetric}} \tag{3.31}$$

All four terms are invariant under simultaneous inversion of both momenta, $(\mathbf{k}, \mathbf{k}') \to (-\mathbf{k}, -\mathbf{k}')$. Instead, if we invert just one momentum of the pair, the first two are symmetric for both arguments $\mathbf{k}$, $\mathbf{k}'$; the second two are anti-symmetric. Decoupling the symmetric part, we obtain

$$c_x c'_x + c_y c'_y = \frac{1}{2}(c_x + c_y)(c'_x + c'_y) + \frac{1}{2}(c_x - c_y)(c'_x - c'_y)$$

which finally gives:

$$\begin{aligned}
\cos(\delta k_x) + \cos(\delta k_y) = {} & \frac{1}{2}(c_x + c_y)(c'_x + c'_y) && (s^*\text{-wave}) \\
& + s_x s'_x && (p_x\text{-wave}) \\
& + s_y s'_y && (p_y\text{-wave}) \\
& + \frac{1}{2}(c_x - c_y)(c'_x - c'_y) && (d_{x^2-y^2}\text{-wave})
\end{aligned}$$

or, in a more compact form

$$\cos(\delta k_x) + \cos(\delta k_y) = \frac{1}{2}\sum_\gamma \varphi_{\mathbf{k}}^{(\gamma)} \varphi_{\mathbf{k}'}^{(\gamma)*} \quad \text{for } \gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\} \tag{3.32}$$

$$= \frac{1}{2}\sum_\gamma \varphi_{\mathbf{k}}^{(\gamma)*} \varphi_{\mathbf{k}'}^{(\gamma)} \tag{3.33}$$

This decomposition will result particularly handy in later calculations.

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